Breaking the Curse of Repulsion: Remoteness-Aware Control of Negative Off-Policy Updates

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Abstract

Off-policy policy optimization reuses historical behavior, including negative-advantage samples that suppress known failures. We show that repeated reuse can turn this useful signal into excessive repulsion: as the learner moves away from a historical negative action, subsequent updates make that action increasingly remote without necessarily reducing its update strength. Our aggregate theory characterizes the resulting transition from a stable displacement beyond the positive-only target to persistent drift and the loss of finite stable equilibria; controlled strength sweeps show that an intermediate displacement can improve held-out reward. The relevant learner-relative coordinate is squared standardized distance for Gaussian policies and surprisal for categorical policies. We introduce Dynamic Remoteness-Aware Policy Optimization (DRPO), which leaves the negative update unchanged in the near field and exponentially attenuates its remote tail. DRPO restores eventual inward Gaussian drift for every fixed finite negative-to-positive mass ratio, yields an explicit ultimate-bound radius, and changes categorical support suppression from exponential to polynomial probability decay. External diagnostics and controlled interventions isolate remoteness-dependent policy geometry as a source of negative-update amplification and show that selective tapering can remove its far-field effect without discarding useful local feedback.

Keywords

Off-Policy Reinforcement Learning, Negative Feedback, Policy Optimization, Stability

1 Introduction

Policy-based methods learn from signed feedback: positive updates attract the policy toward actions estimated to be preferable, whereas negative updates suppress actions estimated to be undesirable. Negative feedback can therefore encode useful boundaries and exclusion information that positive-only learning lacks [16, 18]. In off-policy settings, however, the same historical sample may be reused after earlier updates have already changed its probability under the current policy. This raises a single question: when does repeated reuse turn useful negative feedback into excessive repulsion?

We identify learner-relative remoteness as the state variable that links one reuse step to the next. A negative update lowers the probability of its historical action, increasing the action's remoteness and thereby changing the next policy-score response. Informally, we call historical actions below the control threshold the *near field* and those beyond it the *far field*; Section 5 makes this distinction operational through $D \leq \tau$ and $D > \tau$. The policy family determines the resulting tail law: Gaussian mean scores grow with standardized distance, whereas categorical logit scores remain bounded even as selected-action probabilities approach the simplex boundary. Under isolated reuse, uncontrolled Gaussian remoteness grows geometrically and categorical action probability decays exponentially; an exponential taper reduces the corresponding far-field growth to logarithmic order. This is the curse of repulsion: complying with a negative update can make its next reuse more remote without making the reused signal self-limiting.

What remains unresolved is whether remoteness itself amplifies negative updates independently of sample quality. Existing methods control off-policy instability through filtering, importance weighting, probability-aware gates, and tapered updates [1–3, 7, 8, 12–14], but remoteness is usually confounded with advantage magnitude and data quality. We therefore match those factors while varying only remoteness, then intervene separately on near- and far-field negative updates to test their downstream effects.

Our aggregate analysis places these per-sample effects in competition with positive attraction. When positive mass dominates, repulsion can displace the policy beyond the positive-only target while preserving a finite stable equilibrium. At balance, the restoring force disappears and persistent drift emerges; under negative dominance, no finite stable equilibrium remains. These regimes characterize policy geometry and stability, not task utility, which must be determined empirically.

Motivated by this transition, we introduce Dynamic Remoteness-Aware Policy Optimization (DRPO). DRPO recomputes remoteness under the current policy, leaves the negative branch unchanged in the near field, and applies an exponential taper beyond the threshold. The taper dominates every finite-order far-field response, restores eventual inward Gaussian drift for any fixed finite negative-to-positive mass ratio, and makes categorical suppression self-limiting. Section 5.3 states the important boundary of this design: remoteness alone cannot identify remote samples that remain aligned with true task improvement.

The experiments separate existence, identification, transmission, and control. External diagnostics first test whether the predicted pattern appears in realistic neural policies; controlled environments then isolate its source, trace its causal effect on policy behavior, and compare selective tapering against positive-only, global-scaling, and polynomial-tail alternatives.

Our contributions are summarized as follows:

- We characterize how repeated negative reuse changes policy-relative remoteness and derive aggregate regimes for stable extrapolation, persistent drift, and the loss of finite stable equilibria.
- We derive a control-order hierarchy and a selective exponential taper with Gaussian ultimate-boundedness and categorical self-limiting-suppression guarantees.
- We use matched external diagnostics and controlled interventions to separate coefficient magnitude from policy geometry, identify a far-field causal pathway, and test the stability–utility trade-off of selective tapering.

2 Related Work

Learning from Negative or Suboptimal Behavior. A broad line of policy-learning methods distinguishes behavior according to its estimated quality. Advantage-weighted regression, AWAC, implicit Q-learning, critic-regularized regression, and behavior-proximal objectives fit actors more strongly toward actions estimated to be valuable under a learned critic [6, 10, 13, 17, 19]. Positive-filtered or filtering-based variants represent a conservative endpoint, where selected negative or low-quality updates are removed rather than reweighted. At the same time, failed or suboptimal behavior can remain informative: it can suppress competing bad modes, sharpen local decision boundaries, or improve data efficiency [1, 2, 12]. These findings support the view that negative feedback is not merely noise, but also carries information whose usefulness depends on how it is used.

Off-Policy, Stale, and Learner-Relative Updates. This issue becomes especially important under off-policy reuse, where historical behavior remains fixed while the learner continues to change. Off-policy methods commonly control behavior–learner mismatch through importance weighting, clipping, trust regions, replay mechanisms, and behavior regularization [3, 5, 7, 9, 14]. Recent asymmetric or tapered off-policy objectives further make updates depend on current-policy probability or surprisal [1, 8]. These works establish that mismatch and rarity matter; our distinction concerns how learner-relative remoteness is created and amplified by repeated negative reuse. This actor-side view is complementary to conservative offline RL, which primarily addresses value extrapolation and policy-support mismatch.

Tapered Updates and Optimistic Reweighting. TOPR uses an asymmetric tapered importance ratio on negative off-policy updates and provides its own stability analysis [8]. Optimistic or best-case distributional optimization also has a formal precedent in likelihood approximation and policy optimization [11, 15]. Our construction is related but distinct: it reweights a finite measure of negative-update mass according to learner-relative remoteness and may reduce that measure's total mass. It is therefore an analogue of optimistic divergence-ball reweighting, not a standard ambiguity set over normalized probability distributions.

3 Problem Setup

3.1 Off-Policy Signed Actor Updates

We consider policy optimization from a historical update distribution ν over state–action pairs, which may represent an offline dataset, a replay buffer, or trajectories collected by earlier or stale behavior policies. Let $\pi_\theta(a \mid s)$ denote the current policy and let $\widehat{A}(s, a)$ denote the advantage-like signal used by the actor update. During an actor step, the update distribution and the associated advantage estimates are treated as fixed with respect to θ. The resulting empirical actor field is

$$\mathbf{F}(\theta) = \mathbb{E}_{(s,a)\sim\nu}\left[\widehat{A}(s, a)\nabla_\theta \log \pi_\theta(a \mid s)\right], \tag{1}$$

with the corresponding update $\theta_{t+1} = \theta_t + \eta\mathbf{F}(\theta_t)$. Repeated reuse means that the same historical samples and signed feedback are evaluated under an evolving current policy; Section 4 studies the actor-side dynamics induced by this reuse.

To separate favorable and unfavorable feedback, we define

$$\begin{aligned} \widehat{A}^+(s, a) &= \max\{\widehat{A}(s, a), 0\}, \\ \widehat{A}^-(s, a) &= \max\{-\widehat{A}(s, a), 0\}. \end{aligned} \tag{2}$$

The actor field can then be decomposed into positive and negative components:

$$\begin{aligned} \mathbf{F}(\theta) = \mathbb{E}_\nu &\left[\widehat{A}^+(s, a)\nabla_\theta \log \pi_\theta(a \mid s)\right] \\ &- \mathbb{E}_\nu \left[\widehat{A}^-(s, a)\nabla_\theta \log \pi_\theta(a \mid s)\right]. \end{aligned} \tag{3}$$

For a negative sample, the update magnitude factorizes as $\widehat{A}^-(s, a)\|\nabla_\theta \log \pi_\theta(a \mid s)\|$, separating the severity of its estimated disadvantage from its learner-relative policy geometry.

We use *coefficient magnitude* for $\widehat{A}^-$, *score response* for $\|\nabla \log \pi\|$, and *update magnitude* for their product. Section 4.1 defines the squared remoteness gradient as the *reuse loop gain*, while *aggregate force* refers to the expectation or sum of signed updates. We use the broader term *influence* only when these distinctions are immaterial.

All aggregate quantities use the same update base measure ν. We write

$$p = \mathbb{E}_\nu[\widehat{A}^+], \qquad q = \mathbb{E}_\nu[\widehat{A}^-], \qquad \rho = q/p \quad (p > 0). \tag{4}$$

This convention matters when comparing conditional sample fields with population fields: sampling frequency already represented by ν is not applied a second time.

4 Far-Field Dynamics of Historical Policy Updates

We study how learner-relative remoteness shapes signed policy updates when historical behavior is reused. We first characterize, in policy-output coordinates, how the remoteness of a sample determines the strength of its score contribution. We then show how repeated reuse turns this static geometry into a dynamic feedback process: positive samples become self-attenuating as the learner approaches them, whereas negative samples become self-amplifying as the learner moves away. Finally, we analyze how these local effects aggregate into finite stable equilibria, persistent drift, or instability, and derive their specific manifestations in Gaussian and categorical policies.

4.1 Learner-Relative Remoteness and Update Strength

To isolate the action-side geometry, we fix a context s and write u for the policy-output coordinates at that context. For a historical action a, we define its learner-relative remoteness as

$$D_u(a) = -\log \pi_u(a). \tag{5}$$

For a categorical policy, $D_u(a)$ is the surprisal of the selected action. For a Gaussian policy, it equals a scale-dependent constant plus one half of the squared Mahalanobis distance from the policy mean. Thus, $D_u(a)$ measures how unlikely the historical action is under the current learner, rather than defining a metric in the action space.

We measure the corresponding squared score norm by

$$R_u(a) = \|\nabla_u \log \pi_u(a)\|_2^2. \tag{6}$$

A sample with advantage estimate $\widehat{A}(s,a)$ contributes an output-space update of magnitude $|\widehat{A}(s,a)|\sqrt{R_u(a)}$. The following results characterize how learner-relative remoteness controls this update strength.

Proposition 4.1 (Distance-dependent score-function magnitude). *Fix a context and a historical action a.*

For a Gaussian policy with mean μ and fixed covariance $\Sigma \succ 0$, let $C_\Sigma = \frac{d}{2}\log(2\pi) + \frac{1}{2}\log|\Sigma|$ denote the Gaussian normalization constant, which is independent of a and μ. Then learner-relative remoteness satisfies

$$D_\mu(a) - C_\Sigma = \frac{1}{2}(a-\mu)^\top \Sigma^{-1}(a-\mu). \tag{7}$$

Moreover, the squared mean-score norm $R_\mu(a) = \|\nabla_\mu \log \pi_\mu(a)\|_2^2$ is bounded above and below by positive constants times $D_\mu(a) - C_\Sigma$, with constants determined only by Σ. Consequently, as standardized distance increases, the Gaussian mean-score response grows without bound. When $\Sigma = \sigma^2 I$, $R_\mu(a) = 2(D_\mu(a) - C_\Sigma)/\sigma^2$, so remoteness and squared mean-score norm have the same global ordering.

For a categorical policy with logits z, learner-relative remoteness is the selected-action surprisal $D_z(a) = -\log \pi_z(a)$. The selected-logit score response is

$$\left|\frac{\partial}{\partial z_a}\log \pi_z(a)\right| = 1 - e^{-D_z(a)}. \tag{8}$$

Thus, categorical surprisal can diverge as the selected probability approaches zero, but the selected-logit response saturates at one. The full logit-score norm remains bounded as well.

The proof, including the explicit Gaussian eigenvalue bounds and the full categorical score-norm bound, is given in Appendix A.1.

Proposition 4.1 identifies the policy-family split used below. In Gaussian policies, remoteness corresponds to squared standardized distance and can produce unbounded mean-score response. In categorical policies, remoteness is surprisal: it can grow without bound as probability approaches zero, but the logit score remains bounded.

4.2 Self-Amplification under Historical Reuse

The preceding result compares samples at a fixed policy. We now consider a fixed historical action that is repeatedly reused while the learner changes. Write

$$D(u) = D_u(a), \qquad R(u) = \|\nabla_u D(u)\|_2^2,$$

and consider the repeated single-sample update

$$u_{t+1} = u_t + \eta\widehat{A}\nabla_u \log \pi_{u_t}(a) = u_t - \eta\widehat{A}\nabla_u D(u_t). \tag{9}$$

Here, the action a and its signed coefficient $\widehat{A}$ are held fixed over the reuse window.

Theorem 4.2 (Self-attenuation and self-amplification under reuse). *Assume that $D(u)$ is differentiable and convex on a convex neighborhood containing the update segments, and define*

$$D_t = D(u_t), \qquad R_t = R(u_t).$$

If $\widehat{A} < 0$, then every reuse step satisfies

$$D_{t+1} \geq D_t + \eta|\widehat{A}|R_t, \qquad R_{t+1} \geq R_t. \tag{10}$$

Thus, negative reuse makes the historical action increasingly remote while preventing its score-function norm from decreasing.

If $\widehat{A} > 0$, and D additionally has an L-Lipschitz gradient on the same neighborhood with $0 < \eta\widehat{A} \leq 1/L$, then

$$D_{t+1} \leq D_t - \frac{\eta\widehat{A}}{2}R_t, \qquad R_{t+1} \leq R_t. \tag{11}$$

Thus, positive reuse makes the historical action increasingly compatible with the learner while attenuating its subsequent score response.

A proof is provided in Appendix A.2. For the two policy-output coordinates used in the main theory, the required convexity condition holds globally: fixed-covariance Gaussian remoteness is quadratic in the mean, and categorical negative log-likelihood is convex in logits.

Theorem 4.2 separates a one-time distant update from a historical-reuse feedback loop. Under a negative coefficient, the learner moves in the direction that increases the historical action's remoteness; after this movement, repeated reuse preserves or increases the sample's score response. Combined with Proposition 4.1, this yields unbounded amplification for fixed-covariance Gaussian mean updates unless the reused action has zero mean score, and bounded but persistent amplification for categorical logits.

Proposition 4.3 (Quantitative Gaussian reuse rates). *For an isotropic Gaussian mean policy with covariance $\sigma^2 I$, one fixed negative sample of mass $q > 0$, and $X_t = D_t - C_\Sigma > 0$, the uncontrolled update satisfies*

$$X_{t+1} = \left(1 + \eta q/\sigma^2\right)^2 X_t,$$

so remoteness grows geometrically. Anticipating the taper introduced in Section 5, if the same sample is weighted by $\exp\{-a[D_t - \tau]_+\}$, $a > 0$, then its continuous-time remoteness is $O(a^{-1}\log t)$, and the discrete update obeys

$$X_t \leq a^{-1}\{\log t + \log\log t + O(1)\}.$$

The proof is in Appendix A.3. This slowdown does not by itself prove aggregate boundedness; that requires the closed-loop attraction established in Theorem 5.3.

4.3 Aggregate Equilibria under Historical Reuse

The preceding results describe the repeated influence of an individual historical action, but they do not imply that the aggregate policy dynamics must diverge. Positive and negative historical samples exert competing attractive and repulsive forces. We therefore ask whether their aggregate effect admits a finite stable equilibrium or instead produces persistent drift or instability.

To analyze continuous-action Gaussian and discrete-action categorical policies within a single argument, we use a common exponential-family output representation at a fixed context; the exact coordinates and regularity assumptions are given in Appendix A.7. Using the aggregate masses p, q, ρ from Equation (4), let $\mathbf{m}_+$ denote the positive-only target, the mean-parameter point selected by positive feedback alone. When $q > 0$, let $\mathbf{m}_-$ denote the magnitude-weighted moment of negative samples. It is not an attraction target: the negative actor contribution repels the learner from this moment.

Theorem 4.4 (Aggregate attraction–repulsion regimes). *Consider the continuous aggregate actor dynamics and the corresponding discrete update under the regular minimal exponential-family representation specified in Appendix A.7. Assume $p > 0$.*

Positive-only case $(\rho = 0)$**.** *If $\mathbf{m}_+$ lies in the interior of the attainable mean-parameter space, there is a unique finite equilibrium whose mean parameter is $\mathbf{m}_+$. It is locally asymptotically stable in continuous time and for sufficiently small discrete step sizes.*

Positive-dominant regime $(0 < \rho < 1)$**.** *Define the signed target*

$$\mathbf{m}^\star(\rho) = \mathbf{m}_+ + \frac{\rho}{1-\rho}\left(\mathbf{m}_+ - \mathbf{m}_-\right). \tag{12}$$

If $\mathbf{m}^\star(\rho)$ lies in the interior of the attainable mean-parameter space, there is a unique finite equilibrium whose mean parameter is $\mathbf{m}^\star(\rho)$. It is locally asymptotically stable in continuous time and for sufficiently small discrete step sizes. Holding $\mathbf{m}_+$ and $\mathbf{m}_-$ fixed, its displacement from $\mathbf{m}_+$ grows without bound as $\rho \to 1^-$, unless $\mathbf{m}_+ = \mathbf{m}_-$. If the signed target leaves the attainable mean-parameter space, no finite equilibrium exists.

Critical regime $(\rho = 1)$**.** *If $\mathbf{m}_+ \neq \mathbf{m}_-$, the policy-output coordinate exhibits persistent drift and admits no finite equilibrium. If $\mathbf{m}_+ = \mathbf{m}_-$, every point is stationary, but no point is an isolated asymptotically stable equilibrium.*

Negative-dominant regime $(\rho > 1)$**.** *Every finite stationary point, if one exists, is unstable in continuous time and under every discrete update with positive step size. Hence, this regime admits no locally asymptotically stable finite equilibrium.*

The proof, including the exponential-family actor field and the exact discrete-time step-size condition, is provided in Appendix A.7. Equation (12) gives the central geometric interpretation: while positive feedback remains dominant, negative feedback displaces the equilibrium from $\mathbf{m}_+$ in the direction $\mathbf{m}_+ - \mathbf{m}_-$, and therefore away from $\mathbf{m}_-$. For fixed moments, the multiplier $\rho/(1-\rho)$ increases with the relative negative strength and diverges as $\rho \to 1^-$. These conclusions concern policy geometry and stability; they do not imply that the resulting displacement improves task utility.

4.4 Policy-Family Manifestations of Negative Dominance

Theorem 4.4 establishes that negative dominance, $p < q$, admits no finite stable equilibrium. It does not, however, determine how this instability manifests or whether policy-output score norms become unbounded. This depends on the policy family.

Theorem 4.5 (Policy-family score behavior under negative dominance). *Assume $p < q$.*

Gaussian policy. *Consider a Gaussian policy $\pi_\mu = \mathcal{N}(\mu, \Sigma)$ with fixed $\Sigma \succ 0$. Its aggregate mean dynamics have a unique finite stationary point, which is repelling. Unless initialized exactly at that point, the distance from it diverges under both the continuous dynamics and every discrete trajectory with fixed step size $\alpha > 0$. Consequently, for every fixed historical action a,*

$$\left\|\nabla_\mu \log \pi_\mu(a)\right\|_2 = \left\|\Sigma^{-1}(a-\mu)\right\|_2 \longrightarrow \infty. \tag{13}$$

Categorical policy. *Consider a categorical policy $\pi_z = \mathrm{softmax}(z)$ with at least two actions and the zero-sum gauge $\mathbf{1}^\top z = 0$. Any finite stationary point, when one exists, is unique and repelling. Unless initialized exactly at that point, the continuous dynamics and every discrete trajectory with fixed step size $\alpha > 0$ eventually leave every compact subset of the logit space. Along some subsequence, the corresponding probability vector approaches the boundary of the simplex. Nevertheless, for every action a and every finite z,*

$$\begin{aligned}\|\nabla_z \log \pi_z(a)\|_2^2 &= \|e_a - \pi_z\|_2^2 \\ &\le 2\left(1-\pi_z(a)\right)^2 \le 2.\end{aligned} \tag{14}$$

Thus, categorical logits can become unbounded and their probabilities can approach the support boundary, while every individual logit-score norm remains bounded.

The proof is provided in Appendix A.8. Fixed covariance is sufficient for the Gaussian result; learned covariance is not required. Neither policy-family conclusion alone implies task-performance collapse or a NaN/Inf numerical failure.

4.5 Why Selective Tapering Works

The preceding regimes identify the failure; the following radial limit identifies the control property that reverses it.

Lemma 4.6 (Far-field radial balance). *For a fixed-covariance Gaussian field with finite negative atoms,*

$$F(\mu) = p\Sigma^{-1}(m_+ - \mu) + \sum_i q_i \Sigma^{-1}(\mu - a_i),$$

$$q = \sum_i q_i,$$

and $\mu = m_+ + rv$, $\|v\|_2 = 1$,

$$\lim_{r\to\infty} \frac{\langle v, F(\mu)\rangle}{r} = (q-p)v^\top \Sigma^{-1} v.$$

Corollary 4.7 (Selective tapering versus global scaling). *If each negative atom is multiplied by $\omega_i(D_i) \to 0$, the limit in Lemma 4.6 becomes $-pv^\top\Sigma^{-1}v < 0$. Thus, for every fixed finite $\rho = q/p$, without ρ-specific tuning, the far-field radial component is eventually inward; the onset radius is not uniform in ρ and depends on the tail rate of ω. If additionally $\omega(D)\sqrt{D} \to 0$, the absolute tapered negative force vanishes. In contrast, global scaling $\omega \equiv \alpha$ is inward only when $\alpha q < p$. Proofs are in Appendix A.4.*

Proposition 4.8 (Restoration by reference regularization). *The regularized linear field has a unique finite stable equilibrium when $p > q$ for every $\gamma \ge 0$, when $p = q$ for every $\gamma > 0$, and when $p < q$ if and only if $\gamma > q - p$. At $p = q$, its displacement is $O(\gamma^{-1})$. Reference regularization restores stability by tethering the policy; DRPO instead removes the far-field source. The proof and the Euclidean-spring variant are in Appendix A.5.*

Corollary 4.7 provides the mechanism-level justification for Section 5: selective tapering changes the far-field sign by removing the remote source, whereas a constant global scale must be tuned to the negative-to-positive mass ratio. These conclusions assume frozen data, fixed advantages, and unbounded reuse. Data refresh, actor–critic coupling, and reference regularization change the field; the proposition above shows how regularization instead restores stability by tethering the policy.

5 Dynamic Remoteness-Aware Policy Optimization

Section 4.5 identifies the design target: control the remote negative source without uniformly weakening the negative branch. We first define DRPO, then derive its control-order hierarchy and closed-loop guarantees.

5.1 Update and Policy-Family Instantiations

For remoteness D, DRPO uses the thresholded exponential taper

$$\omega_{\mathrm{Exp}}(D) = \exp\left\{-\lambda \left[\frac{D-\tau}{c}\right]_+\right\}, \tag{15}$$

where τ marks the control threshold, $c > 0$ fixes the remoteness scale, and $\lambda > 0$ controls the decay rate. The weight is exactly one when $D \le \tau$ and decays smoothly when $D > \tau$.

Using $D_\theta(s,a) = -\log \pi_\theta(a \mid s)$, define $\bar{D}_\theta(s,a) = \mathrm{sg}[D_\theta(s,a)]$, where $\mathrm{sg}[\cdot]$ denotes stop-gradient. DRPO changes only the negative branch:

$$\widetilde{A}_\theta(s,a) = \widehat{A}^+(s,a) - \omega_{\mathrm{Exp}}\big(\bar{D}_\theta(s,a)\big)\,\widehat{A}^-(s,a), \tag{16}$$

with actor field

$$\mathbf{F}_{\mathrm{DRPO}}(\theta) = \mathbb{E}_{(s,a)\sim\nu}\left[\widetilde{A}_\theta(s,a)\nabla_\theta \log \pi_\theta(a \mid s)\right]. \tag{17}$$

Remoteness is recomputed under the current policy but detached before it is used as a coefficient; DRPO is therefore a dynamic weighting rule rather than a differentiable distance regularizer.

At each policy iterate, tapering preserves the algebraic regime structure of Theorem 4.4 after replacing the negative mass and moment by

$$q_\omega = \mathbb{E}_\nu[\widehat{A}^-\omega], \qquad \mathbf{M}_{-,\omega} = \mathbb{E}_\nu[\widehat{A}^-\omega T(a)].$$

When $0 < q_\omega < p$, the controlled signed target is $\mathbf{m}^\star_\omega = (p\mathbf{m}_+ - \mathbf{M}_{-,\omega})/(p - q_\omega)$; when $q_\omega = 0$, use $\mathbf{M}_{-,\omega} = 0$ without defining a normalized negative moment.

Gaussian policies. For a Gaussian policy with mean $\mu_\theta(s)$ and fixed covariance Σ, define

$$D_{\mathrm{G},\theta}(s,a) = (a - \mu_\theta(s))^\top \Sigma^{-1}(a - \mu_\theta(s)). \tag{18}$$

Since $D_\theta - C_\Sigma = \frac{1}{2}D_{\mathrm{G},\theta}$, the additive constant and factor 1/2 can be absorbed into the threshold and scale:

$$\omega_{\mathrm{G}}(s,a) = \exp\left\{-\lambda_{\mathrm{G}}\left[\frac{D_{\mathrm{G},\theta}(s,a) - \tau_{\mathrm{G}}}{c_{\mathrm{G}}}\right]_+\right\}. \tag{19}$$

Categorical policies. For a categorical policy, substituting selected-action surprisal into Equation (15) gives

$$\omega_{\mathrm{C}}(s,a) = \min\left\{1,\ \left(e^\tau \pi_\theta(a \mid s)\right)^{\lambda/c}\right\}. \tag{20}$$

Thus, the same actor update uses squared standardized distance for Gaussian policies and surprisal for categorical policies.

5.2 Loop-Gain Order

The quantity that drives the next change in remoteness is not the single-step parameter displacement but the squared score $R(D) = \|\nabla D\|_2^2$. We call this the reuse *loop gain.*

Proposition 5.1 (Gaussian loop-gain order). *For a fixed-covariance Gaussian mean policy, $I_{\mathrm{loop}}(D) := R(D) = \Theta(D)$ in the far field. Hence $\omega(D)I_{\mathrm{loop}}(D)$ is bounded exactly at the matching order $\omega(D) = O(D^{-1})$, and vanishes when $\omega(D) = o(D^{-1})$. Equivalently, the matching order is $O(r^{-2})$ in standardized distance r, since $D - C_\Sigma = r^2/2$. The proof is in Appendix A.9.*

At the exact matching rate $\omega(D) = \Theta(D^{-1})$, the weighted parameter force is $\Theta(D^{-1/2}) \to 0$, but the weighted reuse loop gain is $\Theta(1)$. An isolated reused atom can therefore keep driving remoteness outward at a constant order. Quantitatively, uncontrolled Gaussian reuse gives geometric growth, matching-order tapering gives linear growth, and DRPO's exponential taper gives logarithmic growth.

5.3 Distance-Dependent Utility of Negative Feedback

Bounded loop gain alone does not determine whether a remote update is useful. Appendix A.10 gives an empirical utility definition and a sufficient shrinkage result, used here only as motivation; we do not assume a universal monotone utility law in remoteness. When a historical negative no longer supplies local correction information, the design target is

$$\omega(D)I(D) \longrightarrow 0 \qquad \text{as } D \longrightarrow \infty. \tag{21}$$

This is a conditional design principle, not a claim that every remote sample is harmful. If far-field feedback remains aligned with the true task-improving direction, tapering can remove useful signal.

5.4 Guarantee Hierarchy

The DRPO taper satisfies the stronger requirement above for every finite-order response considered here.

Proposition 5.2 (Vanishing finite-order far-field response). *Let $I(D)$ denote the raw far-field response scale. If, for some finite $m \ge 0$ and $C > 0$, $I(D) \le C(1+D)^m$ in the far field, then the taper in Equation (15) satisfies*

$$\omega_{\mathrm{Exp}}(D)I(D) \longrightarrow 0 \qquad \text{as } D \to \infty.$$

Thus, the exponential taper eliminates any finite-order far-field tail rather than merely bounding it.

The exponential form is not unique; it is a simple smooth envelope that dominates finite-order growth. The proof is given in Appendix A.11.

The control conditions form a guarantee hierarchy. L0, $\omega(D) \to 0$, gives eventual inward Gaussian radial drift for each fixed finite ρ; L1, $\omega(D)\sqrt{D} \to 0$, removes the absolute negative far-field force; L2, $\omega(D)D = O(1)$, bounds the Gaussian reuse loop gain; and L3, $D^k\omega(D) \to 0$ for every finite k, removes all finite-order tails. Global scaling fails L0, reciprocal-quadratic tapering is critical at L2, and exponential tapering satisfies L0–L3. This hierarchy ranks guarantees, not task performance.

This hierarchy also organizes the experimental controls. Positive-only and global scaling ignore remoteness; reciprocal tapers have polynomial tails, and the quadratic member meets the matching order; DRPO uses the exponential member that satisfies L0–L3. These controls share the actor form in Equation (17) and differ only in their negative-branch coefficient [1, 4, 6, 8, 10, 13].

5.5 Output-Coordinate Closed-Loop Guarantees

We now record the output-coordinate consequences of the DRPO update once positive attraction is included.

Theorem 5.3 (Gaussian DRPO boundedness). *Let $p > 0$, $\Sigma \succ 0$, and*

$$F_\omega(\mu) = p\Sigma^{-1}(m_+ - \mu) + \sum_i q_i \omega_i(D_\mu(a_i))\Sigma^{-1}(\mu - a_i).$$

For finitely many negative atoms with exponential tapers, define

$$\begin{aligned} B_\omega &= \sup_\mu \left\| \sum_i q_i \omega_i(D_i)\Sigma^{-1}(\mu - a_i) \right\|_2, \\ R^* &= \frac{B_\omega}{p\lambda_{\min}(\Sigma^{-1})}. \end{aligned} \tag{22}$$

Every continuous trajectory satisfies $\limsup_{t\to\infty} \|\mu(t) - m_+\|_2 \le R^*$, and the field has at least one finite equilibrium. If an uncontrolled stable equilibrium places every negative atom strictly below its threshold, the controlled field and Jacobian coincide with the original ones in a neighborhood of that equilibrium.

The proof gives an explicit branch-wise bound on B_ω; a separate branch is required when the shifted threshold is negative because the taper is then active at radius zero. Along every escaping sequence, $\rho_{\text{eff}}(\mu) = p^{-1}\sum_i q_i\omega_i(D_i) \to 0$. This is a diagnostic consequence, not the boundedness argument. See Appendix A.6.

Theorem 5.4 (Categorical self-limiting suppression). *For a repeatedly suppressed action i, define $y_i = z_i - \log\sum_{j\ne i} e^{z_j}$, so $\pi_i = \sigma(y_i)$. Under an uncontrolled negative atom of mass $q > 0$,*

$$\dot{y}_i = -q(1-\pi_i)c_K(t), \qquad \frac{K}{K-1} \le c_K(t) \le 2.$$

Hence y_i eventually decreases linearly and π_i exponentially. If one off-target action is initially the unique maximum then $c_K(t) \to 2$; under complete off-target symmetry, $c_K(t) \equiv K/(K-1)$. With $\omega_i = \min\{1, (e^{\tau_i}\pi_i)^\beta\}$, $\beta = \lambda/c$, the far branch instead satisfies

$$y_i(t) = -\beta^{-1}\log t + O(1), \qquad \pi_i(t) = \Theta(t^{-1/\beta}).$$

Thus tapering changes persistent suppression from exponential to polynomial probability decay; it does not freeze the action probability. The output-coordinate proof is in Appendix A.12; shared-network parameter gradients additionally contain the network Jacobian and cross-sample interference.

Remark 5.5 (Restoring-force balance). A stable finite crossing requires the restoring-minus-suppression field to change sign from positive to negative as the log-odds increase; equivalently, at a hyperbolic crossing, $[S_{\text{tap}} - E]'(y^*) > 0$. Entropy regularization with $\beta \ge 1$ satisfies only the required far-tail ordering in the binary or symmetric one-vs-rest reduction; a full stability result additionally uses the stated phase-line conditions. See Appendix A.13.

5.6 Variational Interpretation

DRPO's exponential attenuation admits an optimistic finite-measure reweighting interpretation analogous to divergence-ball DRO. The analogy is deliberately qualified: unlike standard ambiguity sets over normalized probability measures, the present construction permits attenuation of the total negative-update mass. Let $d\xi^-(z) = \widehat{A}^-(z)d\nu(z)$ and $h(z) = [D(z) - \tau(z)]_+$. Consider

$$\min_{0\le w\le 1} \int wh\, d\xi^- \quad \text{s.t.} \quad \int (w\log w - w + 1)d\xi^- \le \delta. \tag{23}$$

Proposition 5.6 (Finite-measure exponential reweighting). *Let $\delta_0 = \xi^-(\{h > 0\})$. For $0 < \delta < \delta_0$, the constraint in Equation (23) is active and has a unique finite $\kappa > 0$ with $w_\kappa(z) = e^{-h(z)/\kappa}$. At $\delta = \delta_0$, the endpoint is $w_0 = \mathbf{1}\{h = 0\}$; for $\delta > \delta_0$, zero objective is already attainable and no finite one-to-one κ–δ relation remains. The later endpoint $\delta \ge |\xi^-|$ merely makes $w \equiv 0$ feasible. The proof is in Appendix B.1.*

5.7 Relation to TOPR

For sequence policies, token surprisal sums to total trajectory surprisal $-\log\pi_\theta(y \mid x)$; mean-token surprisal divides this quantity by $|y|$. The two coordinates have the same ordering at fixed length but need not agree across variable-length responses. TOPR attenuates negative updates by the truncated ratio $\min\{\pi/\mu, 1\}$.

At the atomic-action or total-trajectory-surprisal level, canonical TOPR's negative coefficient is the $\beta = 1$, sample-specific-threshold member of the taper family. Our mean-token-surprisal Countdown implementation is instead a length-normalized analogue. Applying the same density-ratio coefficient to Gaussian policies is an extension under a common dominating measure, not a claim about the scope of the original TOPR analysis. Algebraically,

$$e^{-[-\log\pi + \log\mu]_+} = \min\{\pi/\mu, 1\}, \tag{24}$$

$$e^{-[-L^{-1}\log\pi + L^{-1}\log\mu]_+} = \min\{\pi/\mu, 1\}^{1/L}. \tag{25}$$

The Gaussian extension assumes a common dominating measure, $\mu(a_i) > 0$, and finite support or integrable tails; using a direct log-density ratio avoids conflating sample thresholds with unequal normalization constants.

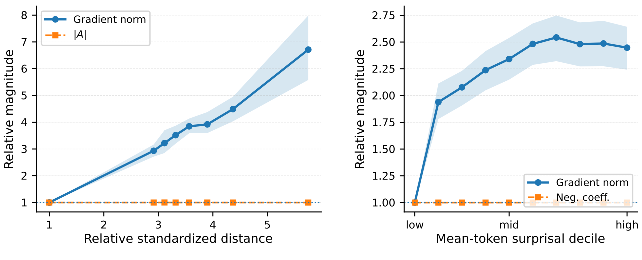
For a fixed reused sample, the far-branch conditional vector field contains a μ^{-1} coefficient, while the resulting absolute probability has a μ prefactor and the normalized ratio π/μ decays as t^{-1}. In the population TOPR field, the outer sampling mass μ cancels this denominator. We therefore report weighted aggregate constants first and label any $\mu_{\min}$ expression as a finite-support worst-case relaxation. In particular, under the applicable plateau branch,

$$B_\omega^{\text{TOPR}} \le \sqrt{\lambda_{\max}(\Sigma^{-1})}\sum_i q_i\sqrt{2\{\log(1/\mu_i) - C_\Sigma\}};$$

otherwise the full branch-wise bound in Appendix A.6 is used.

6 Experiments

The experiments proceed from occurrence to identification, control, and transfer. We first test whether the predicted far-field pattern appears in realistic continuous-control and autoregressive policies. Controlled environments then separate coefficient magnitude from policy geometry, trace the resulting update into policy behavior, and compare negative-weighting rules under known near/far structure. The final task-level sections retain the D4RL and Countdown evaluation targets, including fields whose formal runs have not yet closed.



(a) Hopper

(b) Countdown

Figure 1: External diagnostic of far-field negative-update magnitude. (a) Hopper: relative implemented actor-gradient diagnostic and matched absolute advantage as functions of relative standardized distance for transitions with negative learned-critic advantages. Values are normalized to the near-field reference within each seed. Lines report averages across 10 seeds, and shaded regions denote seed-level bootstrap 95% confidence intervals. (b) Countdown: relative implemented actor-gradient diagnostic and fixed negative coefficient magnitude as functions of current-policy mean-token surprisal deciles among verifier-matched unsuccessful responses. Values are normalized to the lowest-surprisal decile, and shaded regions denote puzzle-level bootstrap 95% confidence intervals.

6.1 Environments and Datasets

We use two controlled environments and two external diagnostic domains. C-U1 is a contextual bandit with a Gaussian policy over continuous actions, while D-U1 uses unordered categorical actions with shared representations. They provide known ground truth for source isolation, policy dynamics, and held-out-context generalization. D4RL Hopper and Countdown arithmetic generation provide continuous offline-control and autoregressive sequence diagnostics, respectively. Full configurations are in Appendix C.

Measurement conventions. When the policy coordinate is controlled, source claims use policy-score response. Neural diagnostics instead report the implemented actor-gradient magnitude only to test whether the policy-level pattern appears in trainable updates; such gradients also contain the network Jacobian.

Outcome taxonomy. We report task-performance collapse, policy-support or variance-boundary events, and NaN/Inf numerical failure as distinct outcome classes. Their protocol-specific definitions are in Appendix D.

6.2 External Diagnostic of Far-Field Repulsion

Hopper provides a continuous-control setting with a state-conditioned Gaussian policy, where remoteness is measured by standardized distance from the current policy. Countdown provides an autoregressive discrete setting, where remoteness is measured by current-policy mean-token surprisal.

The diagnostic controls the sample-quality coefficient in each setting. In Hopper, we compare negative-advantage transitions after matching their absolute advantage magnitudes. In Countdown, we compare verifier-matched unsuccessful responses while holding the negative coefficient magnitude fixed. These matches test whether remoteness is reflected in implemented actor updates independently of the sample-quality coefficient.

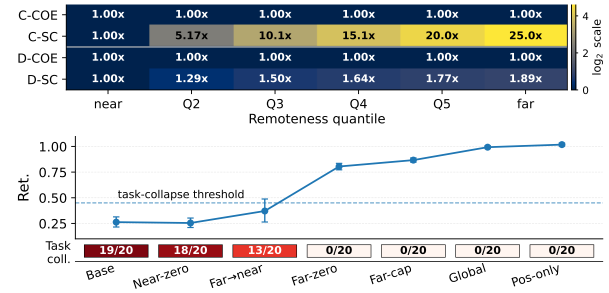


Figure 2: Controlled source isolation and causal transmission. Top: source-isolation factorization across learner-relative remoteness bins, normalized to the near bin. COE denotes coefficient magnitude and SC denotes policy-score response; C rows report the C-U1 continuous protocol, and D rows show the corresponding direct-softmax categorical reference. Bottom: C-U1 causal-intervention summary. Ret. denotes reward retention; points show 20-seed means with bootstrap confidence intervals, and Task coll. reports task-performance collapse counts.

Figure 1 shows signatures specific to each policy family. In Hopper, the implemented actor-gradient diagnostic increases with standardized distance while matched absolute advantage remains approximately flat. In Countdown, the fixed-coefficient diagnostic remains substantial at high mean-token surprisal. We interpret the latter as persistent suppression in a categorical sequence policy, not as an unbounded direct-logit score; implemented parameter gradients also contain the network Jacobian. These diagnostics show that remoteness-correlated update magnitude appears in realistic neural actors, but they do not identify its source or causal effect.

6.3 Controlled Identification of the Causal Mechanism

The external diagnostics still entangle sample quality, remoteness, critic or verifier coefficients, data coverage, and subsequent dynamics. Controlled environments vary coefficient magnitude and remoteness independently and intervene directly on near- and far-field negative updates. Full constructions and protocols are provided in Appendix D.

6.3.1 Remoteness as an Independent Source of Update Magnitude. We first isolate whether remoteness changes update magnitude independently of sample quality. In C-U1, negative actions are placed on the same reward contour, giving them identical fixed advantages but different standardized distances from the current Gaussian policy. The categorical direct-softmax reference applies the same factorization by fixing the negative coefficient while varying current-policy surprisal.

The top panel of Figure 2 separates the coefficient term from the policy-score term. The coefficient magnitude remains flat across remoteness bins, whereas the score response increases with remoteness. Thus, the larger update cannot be explained by farther samples

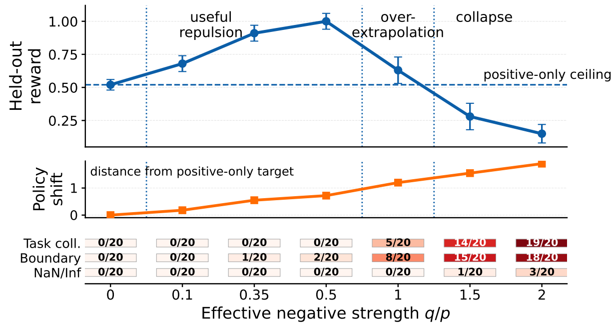
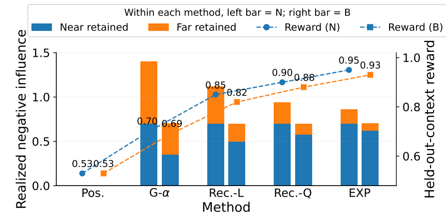


Figure 3: Stable extrapolation from controlled negative feedback. Increasing the effective negative strength q/p first improves held-out-context reward beyond the positive-only ceiling, then causes over-extrapolation and task-performance collapse. Policy shift measures displacement from the positive-only target. The color strip reports task collapse, boundary events, and NaN/Inf failures separately.

(a) Controlled taper comparison



(b) Transfer

Method	Hop.	Cnt.
Pos.	–	–
G-α	–	–
Rec.-L	–	–
Rec.-Q	–	–
DRPO	–	–

Hop.: Hopper return.
Cnt.: Countdown success.

Figure 4: Controlled taper comparison and external transfer. (a) Matched operating-point comparison in the controlled taper protocol. For each method, the left bar is the near-field-retention matched operating point and the right bar is the aggregate-budget matched operating point. Bars decompose retained negative-update magnitude into near- and far-field components, and dashed traces report held-out-context reward. (b) Reserved external validation summary for Hopper and Countdown; dashes mark outcomes awaiting the corresponding formal runs.

having larger advantages; it is generated by the policy-score geometry itself. The manifestation differs across action spaces: Gaussian policies exhibit amplitude amplification, while categorical policies approach a bounded direct-logit score but continue to suppress the selected action toward the simplex boundary. This experiment identifies the source of the update difference, not its downstream causal effect.

6.3.2 Causal Transmission into Drift and Collapse. We next test whether the far-field update causally propagates into instability. In the C-U1 intervention protocol, all branches share the same data, fixed labels, initialization, optimizer, and training horizon, and differ only in how near- and far-field negative updates are retained, removed, capped, or globally rescaled. These matched-control invariants are formalized in Appendix D.1.

The bottom panel of Figure 2 shows a clear rescue pattern. Removing near-field negative updates leaves performance collapse largely unchanged, whereas removing or capping far-field updates restores reward retention and eliminates task-performance collapse. Global rescaling also rescues the run, supporting the interpretation that abnormal negative-update magnitude is a mediator. In contrast, transferring the far-field budget to near-field negatives gives only partial recovery, showing that the rescue is not obtained by simply preserving the same total negative-update budget.

These interventions identify the far-field negative update, rather than negative feedback in general, as a causal transmission path to policy instability in the controlled Gaussian environment. Outcomes follow the taxonomy in Section 6.1.

6.4 From Useful Repulsion to DRPO Control

The causal rescue does not imply that all negative feedback should be removed. C-U1 and D-U1 place a hidden optimum beyond the positive-only target and make local repulsion point toward it, while additional remote negatives create far-field pressure. This construction separates useful repulsion from harmful dominance and permits controlled comparisons of taper shape.

6.4.1 Stable Extrapolation from Controlled Negative Feedback. We sweep the effective negative strength relative to positive attraction, denoted by q/p, under the matched-control invariants above; protocol and metric details are in Appendix D.4. The positive-only setting is $q/p = 0$.

Figure 3 shows a non-monotone transition. For small and moderate q/p, held-out-context reward rises above the positive-only ceiling, showing that controlled negative feedback provides useful repulsion beyond the positive-support solution. As q/p grows larger, policy shift continues to increase but reward declines, indicating over-extrapolation rather than additional useful generalization. At the largest strengths, task-performance collapse and boundary events appear; numerical failures remain a separate outcome class.

This strength sweep supplies the missing link between causal rescue and tapered control: positive-only learning removes the harmful tail but also removes the beneficial intermediate regime. Remoteness-aware control should therefore preserve useful local repulsion while preventing the far-field tail from dominating.

6.4.2 Controlled Comparison of Taper Shapes. We compare positive-only learning, global negative scaling, reciprocal-linear tapering, reciprocal-quadratic tapering, and exponential tapering while changing only the negative weighting rule. Two matched comparisons are used. Near-field-retention matching preserves comparable local negative-update magnitude and isolates differences in far-field decay, whereas aggregate-budget matching tests whether a selective taper outperforms a global reduction with the same total negative-update budget.

As shown in Figure 4, DRPO's exponential taper provides the strongest stability–utility trade-off under both matched comparisons. Positive-only learning avoids far-field instability but remains at the positive-support ceiling. Global scaling suppresses useful near- and harmful far-field feedback indiscriminately. Reciprocal tapering introduces remoteness-aware suppression, but its tail remains heavier than the exponential rule. The exponential taper instead retains near-field repulsion while rapidly suppressing the far-field tail, yielding higher task performance and held-out-context generalization than the matched alternatives.

6.4.3 External Validation of Remoteness Tapering. We finally transfer the same control principle to Hopper and Countdown. DRPO weights Hopper transitions by their standardized remoteness from the current Gaussian policy and Countdown responses by their current mean token surprisal. Within each task, all methods share the dataset, initialization, labels, optimizer, and training budget. Because these external tasks do not provide known counterfactual geometry or exact near-field utility matching, they test transfer rather than repeat the controlled causal identification.

The external summary in Figure 4 pre-registers the transfer comparison, but its cells remain open pending the formal runs and terminal audit. We therefore make no task-level ranking claim from this panel. The controlled results establish the stability–utility mechanism; the external panel tests whether that principle transfers to continuous offline control and autoregressive policy optimization.

6.5 Overall Task Performance

Having established the controlled mechanism, we evaluate DRPO as a complete policy-optimization method on D4RL locomotion and Countdown. We compare it with published domain-specific baselines and protocol-matched controls spanning positive-only learning, global negative scaling, and alternative remoteness tapers. All protocol-matched methods use the same data and compute budgets. We report normalized return for D4RL and verifier success for Countdown, while task-performance failures, support or variance-boundary events, and NaN/Inf numerical failures remain separate outcomes.

6.5.1 D4RL Locomotion. We evaluate the nine D4RL Gym locomotion tasks formed by HalfCheetah, Hopper, and Walker2d under the medium, medium-replay, and medium-expert datasets. These datasets vary in behavior quality and coverage, with medium-replay containing the broadest mixture of historical policies. Table 1 compares DRPO with published offline-RL baselines under the D4RL evaluation protocol. The shared objectives and taper functions are defined in Appendix D.7; D4RL-specific score provenance, uncertainty statistics, remoteness construction, and implementation details are provided in Appendix D.8.

The populated candidate table gives DRPO a D4RL-9 total normalized return of 716.6. This aggregate and the apparent medium-replay advantage remain provisional until the registered seed-level and terminal audits close; no causal explanation is inferred from the task-level ranking.

6.5.2 Countdown. We evaluate DRPO on held-out Countdown puzzles, where a response is successful only if it forms a valid arithmetic expression that uses the provided numbers and reaches the target value. Greedy verifier success is the primary metric, while pass@k measures sampling-level solution coverage and valid-expression rate distinguishes reasoning failure from formatting or execution failure.

All compared policy-optimization methods start from the same SFT checkpoint and use the same frozen response bank, training budget, validation split, and test puzzles. We compare DRPO with AsymRE [1] and TOPR [8] as off-policy baselines. The shared objectives and taper functions are defined in Appendix D.7; Countdown-specific baseline implementations, response-bank construction, checkpoint selection, and evaluation details are provided in Appendix D.9.

Table 2 reserves the greedy success, pass@k, and validity fields for the formal run. Because all entries are pending, we make no Countdown ranking or improvement claim in this version.

7 Conclusion

Repeated off-policy reuse can transform useful negative feedback into a self-reinforcing geometric effect. Our analysis connects the per-sample reuse loop to aggregate regimes of stable displacement, drift, and instability. DRPO interrupts that loop with a detached remoteness taper, yielding Gaussian ultimate boundedness and polynomial rather than exponential categorical suppression. Matched diagnostics and controlled interventions isolate policy geometry as the source of the far-field update and show why selective control can outperform both removing and globally scaling negative feedback. The main open limitation is semantic: remoteness alone cannot distinguish obsolete negative samples from remote samples that remain aligned with task improvement. Combining geometric

Table 1: Normalized returns on the D4RL-9 Gym locomotion benchmark.

Dataset	CQL†	TD3+BC†	IQL†	**DRPO**
halfcheetah-medium	44.0	**48.3**	47.4	47.3
hopper-medium	58.5	59.3	**66.3**	63.3
walker2d-medium	72.5	83.7	78.3	**85.5**
halfcheetah-replay	**45.5**	44.6	44.2	43.3
hopper-replay	95.0	60.9	94.7	**95.8**
walker2d-replay	77.2	**81.8**	73.9	78.1
halfcheetah-expert	**91.6**	90.7	86.7	91.1
hopper-expert	**105.4**	98.0	91.5	101.8
walker2d-expert	108.8	110.1	109.6	**110.4**
D4RL-9 total	*698.5*	*677.4*	*692.4*	***716.6***

Note. Dataset names are shortened for compactness: "replay" denotes "medium-replay", "expert" denotes "medium-expert", and the D4RL "-v2" suffix is omitted. Results marked by † are reported by prior work rather than protocol-matched reruns. Rows report task-level normalized returns, and *D4RL-9 total* sums the nine task means. Bold and underlined values denote the best and second-best results among the methods shown in this table. DRPO results are averaged over 10 runs; standard deviations and seed-level results are reported in Appendix D.8.

Table 2: Performance on held-out Countdown puzzles.

Metric	AsymRE	TOPR	**DRPO**
Greedy success (%)	–	–	–
Pass@k (%)	–	–	–
Valid expression (%)	–	–	–

Note. All methods use the same SFT checkpoint, frozen response bank, validation split, test puzzles, and training budget. A dash denotes a result pending the formal run.

control with a reliable utility or alignment signal is a natural next step.

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A Proofs for the Theoretical Results

Proof map. The score and reuse results are proved in Sections A.1–A.3; the aggregate regimes and policy-family manifestations are proved in Sections A.7 and A.8. The selective-tapering bridge, reference restoration, Gaussian boundedness, control order, exponential tail, and categorical results appear in Sections A.4, A.5, A.6, A.9, A.11, and A.12–A.13, respectively.

A.1 Proof of Proposition 4.1

Proof. Let $\delta = a - \mu$. For a Gaussian policy with fixed covariance Σ, the negative log-likelihood expands as

$$D_\mu(a) = C_\Sigma + \frac{1}{2}\delta^\top \Sigma^{-1}\delta,$$

where $C_\Sigma = \frac{d}{2}\log(2\pi) + \frac{1}{2}\log|\Sigma|$ is independent of a and μ. This proves Equation (7).

The mean score function is

$$\mathbf{s}_\mu(a) = \nabla_\mu \log \pi_\mu(a) = \Sigma^{-1}\delta, \tag{26}$$

and hence

$$R_\mu(a) = \left\|\mathbf{s}_\mu(a)\right\|_2^2 = \delta^\top \Sigma^{-2}\delta. \tag{27}$$

Let $x = \Sigma^{-1/2}\delta$. Then

$$D_\mu(a) - C_\Sigma = \frac{1}{2}\|x\|_2^2, \qquad R_\mu(a) = x^\top \Sigma^{-1} x. \tag{28}$$

Applying the Rayleigh quotient bounds gives

$$\begin{aligned} R_\mu(a) &\geq 2\lambda_{\min}(\Sigma^{-1})\big(D_\mu(a) - C_\Sigma\big), \\ R_\mu(a) &\leq 2\lambda_{\max}(\Sigma^{-1})\big(D_\mu(a) - C_\Sigma\big). \end{aligned} \tag{29}$$

Thus, $R_\mu(a)$ is linearly comparable to $D_\mu(a) - C_\Sigma$, with constants determined only by Σ. Since $D_\mu(a) - C_\Sigma = \frac{1}{2}d^2_{\mathrm{Mah}}(a, \mu)$, the Gaussian mean-score response grows without bound as the standardized distance grows.

When $\Sigma = \sigma^2 I$, Equation (27) gives

$$R_\mu(a) = \frac{1}{\sigma^4}\|\delta\|_2^2, \qquad D_\mu(a) - C_\Sigma = \frac{1}{2\sigma^2}\|\delta\|_2^2.$$

Therefore,

$$R_\mu(a) = \frac{2}{\sigma^2}\big(D_\mu(a) - C_\Sigma\big),$$

which proves the isotropic exact relation stated in the proposition.

For the categorical policy, let

$$p = \pi_z(a) = e^{-D_z(a)}. \tag{30}$$

The score function with respect to the logits is

$$\mathbf{s}_z(a) = \nabla_z \log \pi_z(a) = e_a - \pi_z. \tag{31}$$

Its selected-action component is therefore

$$\left|\frac{\partial}{\partial z_a}\log \pi_z(a)\right| = 1 - p = 1 - e^{-D_z(a)}, \tag{32}$$

which proves Equation (8). Since $1 - e^{-D}$ is strictly increasing for $D \geq 0$ and converges to 1 as $D \to \infty$, the selected-logit response increases with surprisal but remains bounded.

For completeness, the squared norm of the full categorical score function is

$$\begin{aligned} R_z(a) &= \|e_a - \pi_z\|_2^2 \\ &= (1-p)^2 + \sum_{j \neq a} \pi_z(j)^2. \end{aligned} \tag{33}$$

Because the non-selected probabilities sum to $1 - p$, the Cauchy–Schwarz inequality gives

$$\sum_{j \neq a} \pi_z(j)^2 \geq \frac{(1-p)^2}{K-1}. \tag{34}$$

The same sum is at most $(1-p)^2$. Hence,

$$\frac{K}{K-1}(1-p)^2 \le R_z(a) \le 2(1-p)^2. \tag{35}$$

Substituting $p = e^{-D_z(a)}$ shows that the full categorical logit-score norm is bounded, completing the proof. □

A.2 Proof of Theorem 4.2

Before proving the generic reuse result, we verify the convexity condition for the two policy-output coordinates used in the main text. For fixed-covariance Gaussian means,

$$D_\mu(a) = C_\Sigma + \frac{1}{2}(a-\mu)^\top \Sigma^{-1}(a-\mu), \qquad \nabla_\mu^2 D_\mu(a) = \Sigma^{-1} \succeq 0.$$

For categorical logits z, with $p = \mathrm{softmax}(z)$,

$$D_z(a) = \log \sum_j e^{z_j} - z_a, \qquad \nabla_z^2 D_z(a) = \mathrm{diag}(p) - pp^\top \succeq 0,$$

because

$$v^\top(\mathrm{diag}(p) - pp^\top)v = \mathrm{Var}_{i\sim p}(v_i) \ge 0.$$

Proof. Let

$$f(u) = D(u), \qquad g_t = \nabla f(u_t), \qquad R_t = \|g_t\|_2^2.$$

We first consider a negative coefficient $\widehat{A} = -c$ with $c > 0$. Defining $\alpha = \eta c$, the update becomes

$$u_{t+1} = u_t + \alpha g_t.$$

By convexity of f,

$$f(u_{t+1}) \ge f(u_t) + g_t^\top(u_{t+1} - u_t) \tag{36}$$
$$= f(u_t) + \alpha \|g_t\|_2^2. \tag{37}$$

Hence,

$$D_{t+1} \ge D_t + \eta|\widehat{A}|R_t.$$

The gradient of a differentiable convex function is monotone, so

$$(g_{t+1} - g_t)^\top (u_{t+1} - u_t) \ge 0.$$

Using $u_{t+1} - u_t = \alpha g_t$ gives

$$g_{t+1}^\top g_t \ge \|g_t\|_2^2.$$

By the Cauchy–Schwarz inequality,

$$\|g_{t+1}\|_2 \|g_t\|_2 \ge g_{t+1}^\top g_t \ge \|g_t\|_2^2.$$

Whenever $\|g_t\|_2 > 0$, division by $\|g_t\|_2$ yields

$$\|g_{t+1}\|_2 \ge \|g_t\|_2,$$

and therefore $R_{t+1} \ge R_t$. The same result is immediate when $R_t = 0$. The inequality is strict whenever the gradient changes nontrivially along a direction of positive curvature.

We next consider a positive coefficient $\widehat{A} = c$ with $c > 0$. Let $\alpha = \eta c$, so that

$$u_{t+1} = u_t - \alpha g_t.$$

Because f has an L-Lipschitz gradient, the descent lemma gives

$$f(u_{t+1}) \le f(u_t) - \alpha\|g_t\|_2^2 + \frac{L\alpha^2}{2}\|g_t\|_2^2 \tag{38}$$
$$= f(u_t) - \alpha\left(1 - \frac{L\alpha}{2}\right)\|g_t\|_2^2. \tag{39}$$

If $\alpha \le 1/L$, then

$$D_{t+1} \le D_t - \frac{\alpha}{2}R_t = D_t - \frac{\eta\widehat{A}}{2}R_t.$$

It remains to show that the score-function norm does not increase. For a convex function with an L-Lipschitz gradient, the gradient is $1/L$-cocoercive:

$$(g_t - g_{t+1})^\top (u_t - u_{t+1}) \ge \frac{1}{L}\|g_t - g_{t+1}\|_2^2.$$

Let $d = g_{t+1} - g_t$. Since $u_t - u_{t+1} = \alpha g_t$, this implies

$$d^\top g_t \le -\frac{1}{\alpha L}\|d\|_2^2.$$

Consequently,

$$\|g_{t+1}\|_2^2 = \|g_t + d\|_2^2 \tag{40}$$
$$= \|g_t\|_2^2 + 2d^\top g_t + \|d\|_2^2 \tag{41}$$
$$\le \|g_t\|_2^2 + \left(1 - \frac{2}{\alpha L}\right)\|d\|_2^2. \tag{42}$$

Because $\alpha L \le 1$, the final coefficient is nonpositive, and hence

$$R_{t+1} = \|g_{t+1}\|_2^2 \le \|g_t\|_2^2 = R_t.$$

This proves both claims. □

A.3 Quantitative Gaussian Reuse Rates

Proof of Proposition 4.3. Let $\delta_t = \mu_t - a$ and $X_t = \|\delta_t\|_2^2/(2\sigma^2) = D_t - C_\Sigma$. For an uncontrolled negative sample of mass q,

$$\delta_{t+1} = \left(1 + \frac{\eta q}{\sigma^2}\right)\delta_t,$$

which gives the stated exact geometric recurrence.

Now let $\omega_t = e^{-a[D_t - \tau]_+}$. Once $D_t > \tau$, write $b = \eta q/\sigma^2$ and $K_0 = e^{a(\tau - C_\Sigma)}$. Then

$$X_{t+1} = X_t\left(1 + bK_0 e^{-aX_t}\right)^2. \tag{43}$$

Consequently $\Delta X_t \ge 0$, and for all sufficiently large X_t,

$$0 \le \Delta X_t \le C X_t e^{-aX_t} \tag{44}$$

for a finite constant C. Put $Y_t = e^{aX_t}$. Since $aX_t e^{-aX_t} \le 1/e$, Equation (44) also gives a uniform bound $a\Delta X_t \le M$. Hence

$$\Delta Y_t = Y_t(e^{a\Delta X_t} - 1) \tag{45}$$
$$\le C_1 Y_t\, a\Delta X_t \le C_2 \log Y_t. \tag{46}$$

The standard comparison sequence $\overline{Y}_t = Kt\log t$, with K chosen larger than the constant in Equation (46), is eventually a supersolution. Therefore $Y_t = O(t\log t)$ and

$$X_t \le a^{-1}\{\log t + \log\log t + O(1)\}.$$

If an application does not make the sign of ΔX_t explicit, negative increments already satisfy the desired upper comparison, while the nonnegative branch obeys the argument above.

In continuous time, the far-branch equation is $\dot{X} = C_3 X e^{-aX}$. Thus $Y = e^{aX}$ satisfies $\dot{Y} = C_3 \log Y$, and the same comparison gives $X(t) = O(a^{-1}\log t)$. □

A.4 Far-Field Radial Bridge

Proofs of Lemma 4.6 and Corollary 4.7. Set $\mu = m_+ + rv$ with $\|v\|_2 = 1$. Direct expansion gives

$$\frac{\langle v, F(\mu)\rangle}{r} = -p\, v^\top \Sigma^{-1} v + \sum_i q_i v^\top \Sigma^{-1} v \tag{47}$$

$$+ \frac{1}{r}\sum_i q_i \langle v, \Sigma^{-1}(m_+ - a_i)\rangle. \tag{48}$$

The final term vanishes, proving the lemma.

With atom-specific weights, $D_\mu(a_i) \to \infty$ along every ray. For finitely many atoms, $\omega_i(D_i) \to 0$ therefore makes both the weighted coefficient of r and the weighted constant term vanish. The radial limit is $-pv^\top\Sigma^{-1}v$, which is strictly negative. This is a pointwise-in-q/p statement: the radius at which it becomes negative can grow with q/p. Moreover, $\|\Sigma^{-1}(\mu - a_i)\|_2 = O(\sqrt{D_i})$, so $\omega(D)\sqrt{D} \to 0$ removes the absolute negative force. Finally, for $\omega_i \equiv \alpha$, the lemma applies with q replaced by αq, yielding the condition $\alpha q < p$. □

A.5 Reference-Regularized Restoration

Proof of Proposition 4.8. The unweighted Gaussian field with a KL-geometry spring is

$$F_\gamma(\mu) = \Sigma^{-1}\{(q - p - \gamma)\mu + pm_+ - qm_- + \gamma\mu_0\}.$$

Whenever $p - q + \gamma \neq 0$, its unique equilibrium is

$$\mu^*_\gamma = \frac{pm_+ - qm_- + \gamma\mu_0}{p - q + \gamma},$$

and the field Jacobian is $(q - p - \gamma)\Sigma^{-1}$. It is Hurwitz exactly when $\gamma > q - p$. This yields all three regimes in the proposition; for $p = q$, the displayed equilibrium has displacement proportional to $1/\gamma$. When $p > q$, increasing γ contracts the equilibrium toward μ_0, potentially sacrificing extrapolation.

For the Euclidean spring $-\gamma(\mu - \mu_0)$, the Jacobian is $(q - p)\Sigma^{-1} - \gamma I$. In the negative-dominant regime it is Hurwitz exactly when $\gamma > (q - p)\lambda_{\max}(\Sigma^{-1})$; this condition is not interchangeable with the KL-geometry coefficient above. □

A.6 Gaussian Closed-Loop Boundedness

Proof of Theorem 5.3. Let $e = \mu - m_+$ and

$$N_\omega(\mu) = \sum_i q_i \omega_i(D_i)\Sigma^{-1}(\mu - a_i).$$

The finite-atom exponential-taper field is locally Lipschitz and has at most linear growth, so its solutions exist for all forward times. For $V(e) = \frac{1}{2}\|e\|_2^2$,

$$\dot V = -pe^\top\Sigma^{-1}e + e^\top N_\omega(\mu) \tag{49}$$

$$\leq -p\lambda_{\min}(\Sigma^{-1})\|e\|_2^2 + B_\omega\|e\|_2. \tag{50}$$

Thus $\dot V < 0$ outside every ball of radius $R^* + \varepsilon$. The standard ultimate-boundedness argument gives

$$\limsup_{t\to\infty} \|\mu(t) - m_+\|_2 \leq R^*.$$

On the boundary of any ball centered at m_+ with radius greater than R^*, Equation (50) is strictly negative. Continuity of the field and the Poincaré–Bohl inward-pointing theorem therefore imply at least one zero inside the ball. This is an existence result, not a uniqueness or global-convergence claim.

It remains to verify $B_\omega < \infty$ and the stated constant. Put $r_i = \|\Sigma^{-1/2}(\mu - a_i)\|_2$, so $D_i = C_\Sigma + r_i^2/2$, and

$$\|\Sigma^{-1}(\mu - a_i)\|_2 \leq \sqrt{\lambda_{\max}(\Sigma^{-1})}\, r_i.$$

For $a = \lambda/c$ and $\widetilde{\tau}_i = \tau_i - C_\Sigma$, maximize

$$g_i(r) = r\exp\{-a[r^2/2 - \widetilde{\tau}_i]_+\}, \qquad r \geq 0.$$

If $\widetilde{\tau}_i < 0$, the taper is active at zero and the maximum occurs at $r = a^{-1/2}$, giving $a^{-1/2}e^{a\widetilde{\tau}_i - 1/2}$. If $\widetilde{\tau}_i \geq 0$, the untapered plateau ends at $r_\tau = \sqrt{2\widetilde{\tau}_i}$. When $a^{-1/2} \leq r_\tau$, the maximum is r_τ; otherwise it is the same interior-tail value. These are exactly the three displayed branches for G_i, and the triangle inequality proves the bound on B_ω.

For near-field fidelity, let μ^* be an uncontrolled equilibrium with $D_{\mu^*}(a_i) < \tau_i$ for every negative atom. Strictness and continuity provide a neighborhood in which every $\omega_i = 1$. The controlled and uncontrolled fields, including their Jacobians, are therefore identical there.

Finally, along any sequence $\|\mu_n\|_2 \to \infty$, each fixed atom satisfies $D_{\mu_n}(a_i) \to \infty$, hence $\omega_i(D_{\mu_n}(a_i)) \to 0$. Finiteness of the atom set gives $\rho_{\text{eff}}(\mu_n) \to 0$. □

A.7 Derivation and Proof of Theorem 4.4

We use a regular minimal exponential-family representation to cover the continuous-action Gaussian and discrete-action categorical policies in a common set of output coordinates. At a fixed context, write

$$\pi_\xi(a) = h(a)\exp\left\{\xi^\top T(a) - \psi(\xi)\right\}, \qquad \xi \in \mathcal{H}, \tag{51}$$

where ξ is the natural parameter, $T(a)$ is the sufficient statistic, and ψ is the log-partition function. Let

$$\mathcal{M} = \{\nabla\psi(\xi) : \xi \in \mathcal{H}\} \tag{52}$$

denote the attainable mean-parameter space.

For a fixed-covariance Gaussian mean policy, $\xi = \Sigma^{-1}\mu, T(a) = a$, and $\nabla\psi(\xi) = \mu$. For a categorical policy, fixing a reference-logit gauge gives a minimal $(K-1)$-dimensional representation in which $T(a)$ is the corresponding reduced action indicator and $\nabla\psi(\xi)$ is the reduced probability vector. Thus, the two cases differ in their sufficient statistics and attainable mean spaces, but share the derivation below.

Write

$$\widehat{A} = \widehat{A}^+ - \widehat{A}^-, \qquad \widehat{A}^+ = \max\{\widehat{A}, 0\}, \qquad \widehat{A}^- = \max\{-\widehat{A}, 0\}.$$

Define the aggregate positive and negative update masses and their unnormalized moments:

$$p = \mathbb{E}_\nu[\widehat{A}^+(a)], \qquad q = \mathbb{E}_\nu[\widehat{A}^-(a)],$$
$$\bar{\mathbf{m}}_+ = \mathbb{E}_\nu\left[\widehat{A}^+(a)T(a)\right], \tag{53}$$
$$\bar{\mathbf{m}}_- = \mathbb{E}_\nu\left[\widehat{A}^-(a)T(a)\right].$$

When $p > 0$, define

$$\mathbf{m}_+ = \frac{\bar{\mathbf{m}}_+}{p}. \tag{54}$$

This is the positive-only target used in the main text: it is the mean-parameter point selected by positive feedback alone, provided that it lies in $\mathcal{M}$. When $q > 0$, define

$$\mathbf{m}_- = \frac{\bar{\mathbf{m}}_-}{q}. \tag{55}$$

This is the negative-feedback moment, namely the advantage-weighted center of the negative feedback. It is not an attraction target; the negative actor contribution repels the learner from this moment. The unnormalized quantities $\bar{\mathbf{m}}_+$ and $\bar{\mathbf{m}}_-$ remain well defined in the boundary cases $p = 0$ or $q = 0$.

The signed population objective is

$$\mathcal{L}(\xi) = \mathbb{E}_\nu\left[\widehat{A}(a)\log \pi_\xi(a)\right]. \tag{56}$$

Using Equation (51) and $\widehat{A} = \widehat{A}^+ - \widehat{A}^-$, we obtain

$$\mathcal{L}(\xi) = (\bar{\mathbf{m}}_+ - \bar{\mathbf{m}}_-)^\top \xi - (p-q)\psi(\xi) + C, \tag{57}$$

where

$$C = \mathbb{E}_\nu\left[\widehat{A}(a)\log h(a)\right]$$

is independent of ξ. Therefore, the aggregate actor field is

$$\mathbf{F}(\xi) = \nabla\mathcal{L}(\xi) = \bar{\mathbf{m}}_+ - \bar{\mathbf{m}}_- - (p-q)\nabla\psi(\xi), \tag{58}$$

and

$$\nabla^2\mathcal{L}(\xi) = -(p-q)\nabla^2\psi(\xi). \tag{59}$$

Regularity and minimality imply that $\nabla^2\psi(\xi) \succ 0$ throughout $\mathcal{H}$, and that $\nabla\psi$ is one-to-one from $\mathcal{H}$ onto $\mathcal{M}$.

We analyze the continuous dynamics and discrete update

$$\dot{\xi} = \mathbf{F}(\xi), \qquad \xi_{t+1} = \xi_t + \alpha\mathbf{F}(\xi_t), \qquad \alpha > 0. \tag{60}$$

Proof. **Positive-only case.** Suppose $p > 0$ and $q = 0$. Then $\bar{\mathbf{m}}_- = \mathbf{0}$, and Equation (58) becomes

$$\mathbf{F}(\xi) = p\left(\mathbf{m}_+ - \nabla\psi(\xi)\right). \tag{61}$$

A finite equilibrium must therefore satisfy

$$\nabla\psi(\xi^\star) = \mathbf{m}_+. \tag{62}$$

If $\mathbf{m}_+ \in \mathcal{M}$, existence follows because $\nabla\psi$ maps $\mathcal{H}$ onto $\mathcal{M}$, and uniqueness follows because this map is one-to-one.

The continuous field Jacobian is

$$J_{\mathbf{F}}(\xi^\star) = -p\nabla^2\psi(\xi^\star) \prec 0, \tag{63}$$

so the equilibrium is locally asymptotically stable. For the discrete update, the local Jacobian is

$$J_{\text{disc}}(\xi^\star) = I - \alpha p\nabla^2\psi(\xi^\star). \tag{64}$$

Its eigenvalues have magnitude smaller than one whenever

$$0 < \alpha < \frac{2}{p\lambda_{\max}\left(\nabla^2\psi(\xi^\star)\right)}. \tag{65}$$

Hence, the positive-only equilibrium is locally stable under sufficiently small discrete steps. If $\mathbf{m}_+ \notin \mathcal{M}$, no finite equilibrium exists.

Positive-dominant regime. Suppose $p > q > 0$. Since both normalized moments are defined, a finite equilibrium must satisfy

$$\mathbf{F}(\xi^\star) = \mathbf{0}, \tag{66}$$

or equivalently

$$\nabla\psi(\xi^\star) = \frac{\bar{\mathbf{m}}_+ - \bar{\mathbf{m}}_-}{p-q} = \frac{p\mathbf{m}_+ - q\mathbf{m}_-}{p-q} = \mathbf{m}^\star. \tag{67}$$

If $\mathbf{m}^\star \in \mathcal{M}$, existence and uniqueness again follow from the bijectivity of $\nabla\psi : \mathcal{H} \to \mathcal{M}$.

Since $p - q > 0$, Equation (59) gives

$$\nabla^2\mathcal{L}(\xi) \prec 0. \tag{68}$$

Thus, $\mathcal{L}$ is strictly concave, and $\xi^\star$ is its unique finite maximizer.

For the continuous dynamics, the field Jacobian at the equilibrium is

$$J_{\mathbf{F}}(\xi^\star) = -(p-q)\nabla^2\psi(\xi^\star) \prec 0. \tag{69}$$

All eigenvalues are strictly negative, so $\xi^\star$ is locally asymptotically stable.

For the discrete update, the local Jacobian is

$$J_{\text{disc}}(\xi^\star) = I - \alpha(p-q)\nabla^2\psi(\xi^\star). \tag{70}$$

Let $\lambda_i > 0$ denote the eigenvalues of $\nabla^2\psi(\xi^\star)$. The corresponding eigenvalues of $J_{\text{disc}}(\xi^\star)$ are $1 - \alpha(p-q)\lambda_i$. Their magnitudes are strictly smaller than one whenever

$$0 < \alpha < \frac{2}{(p-q)\lambda_{\max}\left(\nabla^2\psi(\xi^\star)\right)}. \tag{71}$$

This proves local discrete-time stability for sufficiently small α.

Define $\rho = q/p$. Algebraically,

$$\mathbf{m}^\star = \frac{p\mathbf{m}_+ - q\mathbf{m}_-}{p-q} = \mathbf{m}_+ + \frac{q}{p-q}\left(\mathbf{m}_+ - \mathbf{m}_-\right) = \mathbf{m}_+ + \frac{\rho}{1-\rho}\left(\mathbf{m}_+ - \mathbf{m}_-\right). \tag{72}$$

This proves Equation (12). In particular, holding $\mathbf{m}_+$ and $\mathbf{m}_-$ fixed, the displacement from the positive-only target is

$$\mathbf{m}^\star - \mathbf{m}_+ = \frac{\rho}{1-\rho}\left(\mathbf{m}_+ - \mathbf{m}_-\right), \tag{73}$$

whose magnitude increases with $\rho \in (0,1)$ whenever $\mathbf{m}_+ \neq \mathbf{m}_-$.

If $\mathbf{m}^\star \notin \mathcal{M}$, no finite $\xi^\star \in \mathcal{H}$ can satisfy Equation (67), and hence no finite equilibrium exists.

More generally, consider any sequence $\mathbf{m}_n^\star \in \mathcal{M}$ that leaves every compact subset of $\mathcal{M}$, either by approaching its boundary or by becoming unbounded, and define

$$\xi_n^\star = (\nabla\psi)^{-1}\left(\mathbf{m}_n^\star\right). \tag{74}$$

Suppose, for contradiction, that $\{\xi_n^\star\}$ remains in a compact subset $K \subset \mathcal{H}$. By continuity, $\nabla\psi(K)$ is a compact subset of $\mathcal{M}$, but $\mathbf{m}_n^\star = \nabla\psi(\xi_n^\star)$ would then remain inside that compact set. This contradicts the assumed behavior of $\mathbf{m}_n^\star$. Hence, the corresponding natural parameters must leave every compact subset of $\mathcal{H}$.

Critical regime. Suppose $p = q > 0$. Equation (58) reduces to

$$\mathbf{F}(\xi) = \bar{\mathbf{m}}_+ - \bar{\mathbf{m}}_- = p\left(\mathbf{m}_+ - \mathbf{m}_-\right), \tag{75}$$

which is independent of ξ.

If $\mathbf{m}_+ \neq \mathbf{m}_-$, the continuous trajectory is

$$\xi(t) = \xi(0) + tp\left(\mathbf{m}_+ - \mathbf{m}_-\right), \tag{76}$$

and the discrete trajectory is

$$\xi_t = \xi_0 + t\alpha p\left(\mathbf{m}_+ - \mathbf{m}_-\right). \tag{77}$$

Both exhibit persistent linear drift and admit no finite equilibrium.

If $\mathbf{m}_+ = \mathbf{m}_-$, then $\mathbf{F}(\xi) = \mathbf{0}$ for every ξ. Every point is stationary, but a perturbed trajectory remains at the perturbed point rather than returning to the original one. Hence, no point is an isolated locally asymptotically stable equilibrium.

Negative-dominant regime. Suppose $p < q$. Equation (59) gives

$$\nabla^2\mathcal{L}(\xi) = (q-p)\nabla^2\psi(\xi) \succ 0. \tag{78}$$

Thus, $\mathcal{L}$ is strictly convex.

If a finite stationary point $\xi^\star$ exists, the continuous field Jacobian is

$$J_{\mathrm{F}}(\xi^\star) = (q-p)\nabla^2\psi(\xi^\star) \succ 0. \tag{79}$$

All eigenvalues are positive, so the stationary point is repelling and therefore unstable.

For the discrete update, the local Jacobian is

$$J_{\mathrm{disc}}(\xi^\star) = I + \alpha(q-p)\nabla^2\psi(\xi^\star). \tag{80}$$

Its eigenvalues are

$$1 + \alpha(q-p)\lambda_i > 1$$

for every $\alpha > 0$ and every $\lambda_i > 0$. Therefore, every finite stationary point is unstable under the discrete dynamics as well. If no stationary point exists, there is trivially no finite stable equilibrium.

The four cases establish Theorem 4.4. ◻

A.8 Proof of Theorem 4.5

Proof. Let

$$r = q - p > 0. \tag{81}$$

Gaussian policy. For a Gaussian policy with fixed covariance,

$$\nabla_\mu \log \pi_\mu(a) = \Sigma^{-1}(a-\mu). \tag{82}$$

The aggregate mean field is therefore

$$\begin{aligned}\mathbf{F}_\mu(\mu) &= \mathbb{E}_\nu\left[\widehat{A}(a)\Sigma^{-1}(a-\mu)\right] \\ &= \Sigma^{-1}\left[p\mu_+ - q\mu_- - (p-q)\mu\right].\end{aligned} \tag{83}$$

Because $p - q = -r$ and

$$\mu^\dagger = \frac{q\mu_- - p\mu_+}{r}, \tag{84}$$

Equation (83) becomes

$$\mathbf{F}_\mu(\mu) = r\Sigma^{-1}\left(\mu - \mu^\dagger\right). \tag{85}$$

For the continuous dynamics, define

$$\delta(t) = \mu(t) - \mu^\dagger. \tag{86}$$

Then

$$\dot{\delta}(t) = r\Sigma^{-1}\delta(t), \tag{87}$$

and hence

$$\delta(t) = \exp\left(r\Sigma^{-1}t\right)\delta(0). \tag{88}$$

Since $\Sigma^{-1} \succ 0$,

$$\|\delta(t)\|_2 \geq \exp\left(r\lambda_{\min}\left(\Sigma^{-1}\right)t\right)\|\delta(0)\|_2. \tag{89}$$

Therefore, unless $\delta(0) = 0$, the distance from the stationary point grows exponentially and $\|\mu(t)\|_2 \to \infty$.

For the discrete update,

$$\mu_{t+1} = \mu_t + \alpha\mathbf{F}_\mu(\mu_t), \tag{90}$$

so

$$\delta_{t+1} = \left(I + \alpha r\Sigma^{-1}\right)\delta_t. \tag{91}$$

Iterating gives

$$\delta_t = \left(I + \alpha r\Sigma^{-1}\right)^t\delta_0. \tag{92}$$

All eigenvalues of the update matrix are strictly larger than one. Consequently,

$$\|\delta_t\|_2 \geq \left[1 + \alpha r\lambda_{\min}\left(\Sigma^{-1}\right)\right]^t\|\delta_0\|_2. \tag{93}$$

Thus, every nonstationary discrete trajectory also diverges.

For any fixed historical action a,

$$\begin{aligned}\left\|\nabla_\mu \log \pi_\mu(a)\right\|_2 &= \left\|\Sigma^{-1}(a-\mu)\right\|_2 \\ &\geq \lambda_{\min}\left(\Sigma^{-1}\right)\|a-\mu\|_2.\end{aligned} \tag{94}$$

Since $\|\mu\|_2 \to \infty$ while a is fixed, $\|a-\mu\|_2 \to \infty$. Equation (94) therefore proves Equation (13).

Categorical policy. Let there be $K \geq 2$ actions and write

$$\pi_z = \mathrm{softmax}(z). \tag{95}$$

Because adding a constant to every logit does not change the policy, we work on the gauge-fixed subspace

$$\mathcal{H}_0 = \left\{z \in \mathbb{R}^K : \mathbf{1}^\top z = 0\right\}. \tag{96}$$

Let ρ_+ and ρ_- denote the normalized positive and negative weighted action distributions, and define

$$b = p\rho_+ - q\rho_-. \tag{97}$$

The signed categorical objective can be written as

$$\mathcal{L}(z) = b^\top z + r\log\left(\sum_{j=1}^K e^{z_j}\right) + C, \tag{98}$$

because $p - q = -r$. Its gradient and Hessian are

$$\nabla\mathcal{L}(z) = b + r\pi_z \tag{99}$$

and

$$\nabla^2\mathcal{L}(z) = r\left[\mathrm{Diag}(\pi_z) - \pi_z\pi_z^\top\right]. \tag{100}$$

For every nonzero $v \in \mathcal{H}_0$,

$$\begin{aligned}v^\top\nabla^2\mathcal{L}(z)v &= r\,\mathrm{Var}_{A\sim\pi_z}\left[v_A\right] \\ &> 0,\end{aligned} \tag{101}$$

because every softmax probability is positive and a nonzero vector in $\mathcal{H}_0$ cannot be constant. Hence, $\mathcal{L}$ is strictly convex on $\mathcal{H}_0$ and has at most one finite stationary point.

Consider first the continuous gradient flow

$$\dot{z} = \nabla\mathcal{L}(z). \tag{102}$$

Along every trajectory,

$$\frac{d}{dt}\mathcal{L}(z(t)) = \|\nabla\mathcal{L}(z(t))\|_2^2 \geq 0. \tag{103}$$

Suppose a nonstationary trajectory were bounded. Its closure would be compact, so $\mathcal{L}$ would remain bounded above. Integrating Equation (103) would then give

$$\int_0^\infty \|\nabla\mathcal{L}(z(t))\|_2^2\,dt < \infty. \tag{104}$$

Since the gradient is Lipschitz on the compact trajectory closure, the standard gradient-flow argument implies

$$\nabla\mathcal{L}(z(t)) \longrightarrow 0. \tag{105}$$

Any accumulation point would therefore be a stationary point.

If no finite stationary point exists, this is an immediate contradiction. If a stationary point $z^\dagger$ exists, strict convexity gives, for every $z \neq z^\dagger$,

$$(z - z^\dagger)^\top \left[\nabla\mathcal{L}(z) - \nabla\mathcal{L}(z^\dagger)\right] > 0. \tag{106}$$

Since $\nabla\mathcal{L}(z^\dagger) = 0$,

$$\frac{d}{dt}\frac{1}{2}\left\|z(t) - z^\dagger\right\|_2^2 > 0 \tag{107}$$

along every nonstationary trajectory. Such a trajectory cannot have a subsequence converging to $z^\dagger$. Hence, the only bounded continuous trajectory is the stationary one itself.

For the discrete update

$$z_{t+1} = z_t + \alpha\nabla\mathcal{L}(z_t), \qquad \alpha > 0, \tag{108}$$

convexity gives

$$\begin{aligned}\mathcal{L}(z_{t+1}) &\geq \mathcal{L}(z_t) + \nabla\mathcal{L}(z_t)^\top(z_{t+1} - z_t)\\ &= \mathcal{L}(z_t) + \alpha\,\|\nabla\mathcal{L}(z_t)\|_2^2.\end{aligned} \tag{109}$$

If the trajectory were bounded, Equation (109) would imply

$$\nabla\mathcal{L}(z_t) \longrightarrow 0. \tag{110}$$

A convergent subsequence would therefore approach a finite stationary point. If no such point exists, this is impossible. If the unique stationary point $z^\dagger$ exists, then for $z_t \neq z^\dagger$,

$$\begin{aligned}\left\|z_{t+1} - z^\dagger\right\|_2^2 &= \left\|z_t - z^\dagger\right\|_2^2\\ &\quad + 2\alpha(z_t - z^\dagger)^\top\nabla\mathcal{L}(z_t)\\ &\quad + \alpha^2\,\|\nabla\mathcal{L}(z_t)\|_2^2\\ &> \left\|z_t - z^\dagger\right\|_2^2.\end{aligned} \tag{111}$$

The trajectory therefore cannot have a subsequence converging to $z^\dagger$. Thus, every nonstationary discrete trajectory leaves every compact subset of $\mathcal{H}_0$.

To relate logit divergence to the probability boundary, note that on $\mathcal{H}_0$ the inverse softmax representation is

$$z_i = \log\pi_z(i) - \frac{1}{K}\sum_{j=1}^{K}\log\pi_z(j). \tag{112}$$

If the probabilities remained in a compact subset of the simplex interior, all coordinates would be bounded below by some $\varepsilon > 0$, and Equation (112) would imply bounded logits. Therefore, an unbounded gauge-fixed logit trajectory must contain a subsequence along which at least one action probability approaches zero.

Finally, for every action a,

$$\nabla_z\log\pi_z(a) = e_a - \pi_z. \tag{113}$$

Its squared norm satisfies

$$\begin{aligned}\|e_a - \pi_z\|_2^2 &= (1 - \pi_z(a))^2 + \sum_{j\neq a}\pi_z(j)^2\\ &\leq (1 - \pi_z(a))^2 + \left(\sum_{j\neq a}\pi_z(j)\right)^2\\ &= 2\,(1 - \pi_z(a))^2\\ &\leq 2.\end{aligned} \tag{114}$$

This proves the categorical score bound and completes the proof. □

A.9 Proof of Minimum-Order Control

For fixed-covariance Gaussian mean coordinates, Proposition 4.1 gives $R(D) = \Theta(D - C_\Sigma) = \Theta(D)$ in the far field. Thus there are constants $0 < c_1 \leq c_2 < \infty$ and D_0 such that, for $D \geq D_0$,

$$c_1 D \leq R(D) \leq c_2 D.$$

If $\omega(D)R(D)$ is bounded, then for some $M < \infty$,

$$\omega(D)R(D) \leq M \qquad \text{for all sufficiently large } D.$$

Using the lower bound gives

$$\omega(D) \leq \frac{M}{c_1 D} = O(D^{-1}).$$

Conversely, $\omega(D) \leq CD^{-1}$ and the upper bound imply

$$\omega(D)R(D) \leq CD^{-1}c_2 D = O(1).$$

The same comparison with little-o proves that the weighted loop gain vanishes exactly when $\omega(D) = o(D^{-1})$. Finally, $D - C_\Sigma = r^2/2$ converts D^{-1} to the equivalent reciprocal-quadratic order r^{-2}.

A.10 Empirical Negative-Update Utility

To operationalize the utility view used in Section 5.3, let $\widehat{J}_{\mathcal{B}}(\theta)$ denote an empirical estimate of task performance on an evaluation batch $\mathcal{B}$. For a negative sample $z = (s, a)$, define its raw repulsive update as

$$g_\theta^-(z) = -\widehat{A}^-(z)\nabla_\theta\log\pi_\theta(a \mid s). \tag{115}$$

Assigning it a weight $w \in [0, 1]$ produces the parameter step

$$\Delta\theta(z; w) = \eta w g_\theta^-(z). \tag{116}$$

We define the empirical utility of this weighted negative update as

$$\widehat{\mathcal{U}}_{\mathcal{B}}(z; w \mid \theta) = \widehat{J}_{\mathcal{B}}(\theta + \Delta\theta(z; w)) - \widehat{J}_{\mathcal{B}}(\theta). \tag{117}$$

The corresponding utility-optimal weight is

$$\widehat{w}^\star_{\mathcal{U}}(z \mid \theta) \in \arg\max_{w\in[0,1]}\widehat{\mathcal{U}}_{\mathcal{B}}(z; w \mid \theta). \tag{118}$$

We distinguish this empirical utility from its population counterpart:

$$\mathcal{U}_\theta(z; w) = J\big(\theta + \eta w g_\theta^-(z)\big) - J(\theta). \tag{119}$$

If $\widehat{J}_{\mathcal{B}}$ is an unbiased estimator of J, then

$$\mathbb{E}_{\mathcal{B}}\left[\widehat{\mathcal{U}}_{\mathcal{B}}(z; w \mid \theta)\right] = \mathcal{U}_\theta(z; w). \tag{120}$$

Thus, empirical utility is a noisy, evaluation-dependent estimate rather than an intrinsic deterministic label attached to the sample.

The following result gives sufficient conditions under which the utility-optimal effective step vanishes in the far field.

Lemma A.1 (Far-field shrinkage of the utility-optimal step). *Let D index learner-relative remoteness, and let $g(D) \neq 0$ denote the corresponding negative-update direction. Write*

$$I(D) = \|g(D)\|_2, \qquad v(D) = \frac{g(D)}{I(D)}. \tag{121}$$

Assume that the local update space admits an orthogonal decomposition

$$\mathcal{T} = \mathcal{T}_{\text{task}} \oplus \mathcal{T}_{\text{far}}, \tag{122}$$

and let P_{task} denote the orthogonal projection onto $\mathcal{T}_{\text{task}}$. Suppose that:

(1) $\nabla J(\theta) \in \mathcal{T}_{\text{task}}$;
(2) *the normalized far-field update becomes asymptotically task-orthogonal,*

$$\|P_{\text{task}} v(D)\|_2 \longrightarrow 0; \tag{123}$$

(3) *there exist $\rho_0 > 0$, $\kappa > 0$, and D_0 such that, for every $D \geq D_0$ and every $\rho \in [0, \rho_0]$,*

$$v(D)^\top \nabla^2 J(\theta + \rho v(D))\, v(D) \leq -\kappa. \tag{124}$$

Define the population utility of an effective step of length ρ by

$$\mathcal{V}_D(\rho) = J(\theta + \rho v(D)) - J(\theta), \qquad 0 \leq \rho \leq \rho_0, \tag{125}$$

and let

$$\rho^\star_{\mathcal{U}}(D) \in \arg\max_{\rho \in [0,\rho_0]} \mathcal{V}_D(\rho). \tag{126}$$

Then

$$\rho^\star_{\mathcal{U}}(D) \longrightarrow 0 \qquad \text{as } D \longrightarrow \infty. \tag{127}$$

Moreover, if

$$\langle \nabla J(\theta), v(D) \rangle < 0,$$

then every nonzero local step has negative utility:

$$\mathcal{V}_D(\rho) < 0 \qquad \text{for all } \rho \in (0, \rho_0]. \tag{128}$$

Proof. Define the first-order task alignment

$$a(D) = \langle \nabla J(\theta), v(D) \rangle\,. \tag{129}$$

Because $\nabla J(\theta) \in \mathcal{T}_{\text{task}}$, orthogonality gives

$$\begin{aligned} |a(D)| &= |\langle \nabla J(\theta), P_{\text{task}} v(D) \rangle| \\ &\leq \|\nabla J(\theta)\|_2 \, \|P_{\text{task}} v(D)\|_2\,. \end{aligned} \tag{130}$$

Equation (123) therefore implies

$$a(D) \longrightarrow 0. \tag{131}$$

For any $\rho \in [0, \rho_0]$, Taylor's theorem gives some $\xi \in (0, \rho)$ such that

$$\begin{aligned} \mathcal{V}_D(\rho) &= \rho a(D) \\ &\quad + \frac{\rho^2}{2} v(D)^\top \nabla^2 J(\theta + \xi v(D))\, v(D). \end{aligned} \tag{132}$$

Using Equation (124),

$$\mathcal{V}_D(\rho) \leq \rho a(D) - \frac{\kappa}{2}\rho^2. \tag{133}$$

Since $\mathcal{V}_D(0) = 0$, any utility-maximizing step must attain nonnegative utility. However, whenever

$$\rho > \frac{2[a(D)]_+}{\kappa}, \tag{134}$$

the right-hand side of Equation (133) is strictly negative. Hence,

$$0 \leq \rho^\star_{\mathcal{U}}(D) \leq \frac{2[a(D)]_+}{\kappa}. \tag{135}$$

Together with Equation (131), this proves Equation (127).

If $a(D) < 0$, then for every $\rho > 0$,

$$\rho a(D) - \frac{\kappa}{2}\rho^2 < 0. \tag{136}$$

Equation (133) therefore implies $\mathcal{V}_D(\rho) < 0$ for every nonzero local step, proving Equation (128). □

The lemma is a sufficient mechanism rather than a universal monotonicity claim. It applies when far-field negative updates become increasingly misaligned with the task-improving direction and the task objective is locally harmed along that far-field direction. The curvature condition can represent local losses caused by parameter drift or policy-support contraction, provided those effects reduce the evaluated objective J. If far-field growth remains aligned with $\nabla J(\theta)$, the conclusion need not hold.

Finally, the effective step generated by a sample weight w is

$$\rho(D; w) = \eta w I(D). \tag{137}$$

Consequently, the population analogue of the utility-optimal sample weight must satisfy

$$\eta w^\star_{\mathcal{U}}(D) I(D) = \rho^\star_{\mathcal{U}}(D) \longrightarrow 0. \tag{138}$$

Thus, when the raw far-field response remains nonvanishing or grows, the utility-optimal sample weight itself must shrink so that the weighted response vanishes.

A.11 Proof of Proposition 5.2

Proof. For $D > \tau$,

$$\omega_{\text{Exp}}(D) = \exp\left[-\frac{\lambda}{c}(D - \tau)\right]. \tag{139}$$

Hence,

$$\begin{aligned} \omega_{\text{Exp}}(D) I(D) &\leq C \exp\left[-\frac{\lambda}{c}(D - \tau)\right](1 + D)^m \\ &\longrightarrow 0, \end{aligned} \tag{140}$$

because exponential decay dominates every finite-order polynomial.

The corresponding weighted parameter step is

$$\Delta\theta_{\text{Exp}}(D) = \eta \omega_{\text{Exp}}(D) g^-_\theta(D), \tag{141}$$

and therefore

$$\left\|\Delta\theta_{\text{Exp}}(D)\right\|_2 = \eta \omega_{\text{Exp}}(D) I(D) \longrightarrow 0. \tag{142}$$

If $\widehat{J}_{\mathcal{B}}$ is locally $L_{\mathcal{B}}$-Lipschitz around θ, then for all sufficiently large D,

$$\begin{aligned} \left|\widehat{\mathcal{U}}_{\theta,\mathcal{B}}(z; \omega_{\text{Exp}}(D))\right| &\leq L_{\mathcal{B}} \left\|\Delta\theta_{\text{Exp}}(D)\right\|_2 \\ &\longrightarrow 0. \end{aligned} \tag{143}$$

This proves both claims. □

A.12 Categorical One-vs-Rest Dynamics

Proof of Theorem 5.4. For a single negative atom i, the direct-logit field is

$$\dot{z} = -q(e_i - \pi), \qquad \dot{z}_i = -q(1 - \pi_i), \qquad \dot{z}_j = q\pi_j \quad (j \neq i).$$

Define the normalized off-target probabilities $\bar{\pi}_j = \pi_j/(1-\pi_i)$. Differentiating $y_i = z_i - \log\sum_{j\neq i} e^{z_j}$ gives

$$\dot{y}_i = -q(1-\pi_i) - q\frac{\sum_{j\neq i}\pi_j^2}{1-\pi_i} \tag{144}$$

$$= -q(1-\pi_i)c_K(t), \qquad c_K(t) = 1 + \sum_{j\neq i}\bar{\pi}_j^2. \tag{145}$$

Because $\bar{\pi}$ is a distribution on $K-1$ actions,

$$\frac{1}{K-1} \le \sum_{j\neq i}\bar{\pi}_j^2 \le 1,$$

which proves the coefficient envelope without a concentration assumption.

We next justify entry into the far half-line before using its bounds. The log-odds is strictly decreasing. Until $\pi_i < 1/2$, monotonicity gives $1-\pi_i(t) \ge 1-\pi_i(0) > 0$, so Equation (145) has a strictly negative upper bound. The trajectory therefore crosses $\pi_i = 1/2$ in finite time. Thereafter $1-\pi_i \ge 1/2$, so $y_i \to -\infty$ at a linear rate and $\pi_i = \sigma(y_i)$ decays exponentially.

For the generic coefficient, suppose $j^* \neq i$ is the unique largest off-target logit initially. For every $k \neq i, j^*$,

$$\frac{d}{dt}(z_{j^*} - z_k) = q(\pi_{j^*} - \pi_k) > 0.$$

The initial ordering is therefore preserved. If a gap stayed bounded, then, once $\pi_i \to 0$, maximality of j^* and the positive initial gap would give a positive lower bound on $\pi_{j^*} - \pi_k$, a contradiction. Hence all such gaps diverge, off-target mass concentrates on j^*, and $c_K(t) \to 2$. Under complete off-target symmetry, symmetry is preserved and $\bar{\pi}_j = 1/(K-1)$, giving $c_K(t) \equiv K/(K-1)$.

Now apply the taper $\omega_i = \min\{1, (e^{\tau_i}\pi_i)^\beta\}$. In the far branch, let $U = e^{-\beta y_i}$. Since $\pi_i e^{-y_i} = 1-\pi_i$,

$$\dot{U} = -\beta e^{-\beta y_i}\dot{y}_i \tag{146}$$

$$= \beta q e^{\beta\tau_i} c_K(t)(1-\pi_i)^{\beta+1}. \tag{147}$$

After the finite crossing above, the right-hand side is bounded above and below by positive constants. Thus $U = \Theta(t)$, which is equivalent to

$$y_i(t) = -\beta^{-1}\log t + O(1), \qquad \pi_i(t) = \Theta(t^{-1/\beta}).$$

□

A.13 Categorical Restoring-Force Balance

Write the one-dimensional restored dynamics as

$$\dot{y} = F(y) = E(y) - S_{\text{tap}}(y),$$

where $S_{\text{tap}} > 0$ is downward suppression and $E > 0$ is upward restoration.

Proposition A.2 (Finite restoring crossing). *Assume E and S_{tap} are continuous and their zeros in $[y_L, y_R]$ are isolated. If $F(y_L) > 0$ and $F(y_R) < 0$, then the interval contains a zero at which the phase line has a positive-to-negative crossing. Such a crossing is locally asymptotically stable; if it is hyperbolic, its condition is*

$$F'(y^*) < 0 \iff [S_{\text{tap}} - E]'(y^*) > 0.$$

Proof. The intermediate value theorem gives at least one zero. Because the zeros are isolated on a compact interval, only finitely many occur; as the endpoint signs differ, at least one has positive sign immediately to its left and negative sign immediately to its right. The flow then points toward that zero from both sides, proving phase-line stability. At a hyperbolic crossing this sign change is equivalent to $F'(y^*) < 0$, and the displayed derivative identity follows from $F = E - S_{\text{tap}}$. □

In the binary or symmetric one-vs-rest tail, the tapered suppression has order $e^{\beta y}$, whereas entropy restoration has order $|y|e^y$ as $y \to -\infty$. Hence $\beta \ge 1$ supplies the required far-tail ordering (with the $|y|$ factor breaking the tie at $\beta = 1$), but it does not by itself verify the finite phase-line conditions of Proposition A.2.

B Distributional Reweighting Interpretations

B.1 Finite-Measure I-Divergence Attenuation

Proof of Proposition 5.6. Use the convention $\phi(0) = 1$ for $\phi(w) = w\log w - w + 1$. For an active constraint, the Lagrangian with multiplier $\kappa > 0$ is

$$\mathcal{L}(w,\kappa) = \int \{wh + \kappa\phi(w)\}\, d\xi^- - \kappa\delta.$$

Pointwise stationarity on $0 < w \le 1$ gives $h + \kappa\log w = 0$, hence

$$w_\kappa(z) = e^{-h(z)/\kappa}.$$

For $h = 0$, this equals one; for $h > 0$, it lies strictly between zero and one. Define $\mathcal{D}(\kappa) = \int \phi(w_\kappa)d\xi^-$. Dominated convergence gives

$$\lim_{\kappa\to\infty}\mathcal{D}(\kappa) = 0, \qquad \lim_{\kappa\downarrow 0}\mathcal{D}(\kappa) = \xi^-(\{h > 0\}) = \delta_0.$$

On a set of positive h, increasing κ strictly increases w_κ toward one and strictly decreases $\phi(w_\kappa)$. Thus $\mathcal{D}(\kappa)$ is continuous and strictly decreasing from δ_0 to zero, proving existence and uniqueness for $0 < \delta < \delta_0$.

At $\delta = \delta_0$, the limit is $w_0 = \mathbf{1}\{h = 0\}$, which has zero objective and divergence δ_0. For $\delta > \delta_0$, the same zero-objective point is strictly feasible, so the constraint need not be active and there is no finite one-to-one multiplier relation. Finally, $\int \phi(0)d\xi^- = |\xi^-|$, so $w \equiv 0$ becomes feasible only at the later endpoint $\delta \ge |\xi^-|$. □

This problem derives learner-relative exponential attenuation. It is distinct from quality-based hard selection: both are optimistic reweighting views, but one controls remoteness under a finite negative measure and the other selects a quality region. Neither problem implies the other.

B.2 TOPR Identities and Conditional-versus-Population Constants

For total surprisal and $\beta = 1$, substituting $D_\pi = -\log\pi$ and $\tau = -\log\mu$ gives

$$e^{-[D_\pi - \tau]_+} = e^{-[-\log(\pi/\mu)]_+} = \min\{\pi/\mu, 1\}.$$

Replacing both log probabilities by their length-normalized values raises the ratio to the $1/L$ power, proving Equations (24)–(25).

For a conditionally reused far-branch sample, the coefficient contains π/μ, so the vector-field prefactor contains μ^{-1}. Solving the asymptotic scalar equation yields $\pi(t) \asymp \mu/(Cqt)$ and $\pi(t)/\mu \asymp$

$1/(Cqt)$. In the population field, however, the outer sampling mass gives

$$\mu \min\{\pi/\mu, 1\} = \min\{\pi, \mu\},$$

and cancels that denominator on the far branch. For Gaussian atoms the triangle inequality and the one-atom G_i calculation in Appendix A.6 give weighted constants first. A $\mu_{\min}$ expression follows only after imposing finite support and replacing the weighted terms by their worst case.

C Experimental Details

C.1 Unified Continuous Environment C-U1

C-U1 is the shared controlled continuous environment used by all continuous mechanism experiments. A context $s \in \mathbb{R}^6$ is sampled from $\mathcal{N}(0, I_6)$. For each seed, we independently sample 4096 training contexts and 4096 test contexts from the same distribution. Test contexts never participate in optimization and are used only to measure held-out-context generalization.

State-dependent task geometry. Each context determines a two-dimensional positive-support center $a_+(s)$ and a unit task direction $u(s)$. We define

$$\begin{aligned}[][a_+(s)]_1 &= 0.70 \tanh\big(0.85 s_1 - 0.30 s_2 s_3 + 0.20 \sin(1.6 s_4)\big),\\ [a_+(s)]_2 &= 0.65 \tanh\big(-0.50 s_2 + 0.35 \cos(1.1 s_5) + 0.22 s_1 s_6\big).\end{aligned} \tag{148}$$

The direction angle is

$$\begin{aligned}\varphi(s) = 1.15 \tanh\big(&0.75 s_1 + 0.50 s_3 - 0.30 s_6\big)\\ &+ 0.30 \sin(1.35 s_2),\end{aligned} \tag{149}$$

with

$$u(s) = \begin{bmatrix}\cos\varphi(s)\\ \sin\varphi(s)\end{bmatrix}, \qquad v(s) = \begin{bmatrix}-u_2(s)\\ u_1(s)\end{bmatrix}. \tag{150}$$

The unobserved task optimum and the directionally useful negative anchor are

$$\begin{aligned}a_\star(s) &= a_+(s) + 0.70 u(s),\\ a_-(s) &= a_+(s) - 0.50 u(s).\end{aligned} \tag{151}$$

Positive and negative actions. Each context is associated with four positive actions and eight negative actions. The positive actions lie on a reward contour of radius $r_+ = 0.75$ centered at $a_\star(s)$:

$$a_{\mathrm{pos},k}(s) = a_\star(s) + r_+\big(\cos\phi_k\, u(s) + \sin\phi_k\, v(s)\big), \tag{152}$$

where

$$\begin{aligned}\{\phi_k\}_{k=1}^4 &= \{\pi - \theta_1,\ \pi + \theta_1,\ \pi - \theta_2,\ \pi + \theta_2\},\\ \theta_1 = 0.20, &\qquad \cos\theta_2 = \frac{2(0.70)}{0.75} - \cos\theta_1.\end{aligned} \tag{153}$$

This construction makes the four actions equal-reward while ensuring that their centroid is exactly $a_+(s)$. It also leaves nonzero conditional action spread, preventing deterministic maximum-likelihood variance contraction from being confused with the far-field mechanism.

The eight negative actions lie on a second reward contour of radius $r_- = 1.20$:

$$a_{\mathrm{neg},j}(s) = a_\star(s) + r_-\big(\cos\psi_j\, u(s) + \sin\psi_j\, v(s)\big), \tag{154}$$

with

$$\{\psi_j\}_{j=1}^8 = \left\{\pi,\ \frac{3\pi}{4},\ \frac{\pi}{2},\ \frac{\pi}{4},\ 0,\ -\frac{\pi}{4},\ -\frac{\pi}{2},\ -\frac{3\pi}{4}\right\}. \tag{155}$$

The first negative action equals $a_-(s)$. All eight negative actions have the same distance to $a_\star(s)$ and therefore the same reward and fixed advantage, while having different distances from the current policy.

Reward and fixed advantages. The ground-truth reward is

$$R(s,a) = \exp\left(-\frac{\|a - a_\star(s)\|_2^2}{2(0.75)^2}\right). \tag{156}$$

Advantages are computed once using the fixed baseline 0.40:

$$\widehat{A}(s,a) = R(s,a) - 0.40, \tag{157}$$

and remain frozen throughout optimization. Consequently, all four positive actions have positive advantages, all eight negative actions have negative advantages, and negative-action quality is exactly matched within each context.

Policy parameterization. The policy is an isotropic state-conditioned Gaussian,

$$\pi_\theta(a \mid s) = \mathcal{N}\left(\mu_\theta(s), \sigma_\theta^2(s) I_2\right). \tag{158}$$

Both $\mu_\theta(s)$ and the scalar $\log\sigma_\theta(s)$ are produced by a shared two-layer ReLU MLP with 64 hidden units per layer and separate output heads. The initial standard deviation is 0.60, and the learnable-variance experiments use no artificial variance clamp. The continuous mechanism results reported in the main text use the fixed-variance branch unless explicitly labeled otherwise. A finite crossing of $\log\sigma_\theta(s) < -12$ is recorded as a support or variance-boundary event, whereas nonfinite parameters or outputs are reported separately as NaN/Inf numerical failure.

Training minibatches are sampled by context, and all actions associated with a selected context are retrieved together. Thus, the replicated actions within one context are not treated as independent contexts. Positive and negative objectives are averaged separately before their relative coefficient is applied, preventing the four-to-eight action count ratio from mechanically changing the total negative-update mass.

Shared use across continuous protocols. All C-U1 experiments reuse the same contexts, reward geometry, fixed advantages, and policy architecture. E1 compares equal-advantage negative actions at different learner-relative distances. E2 applies only positive updates and measures the same negative actions as phantom probes. E3 dynamically partitions negative actions using the standardized distance

$$d_\theta(s,a) = \frac{\|a - \mu_\theta(s)\|_2}{\sigma_\theta(s)}, \tag{159}$$

with the prespecified near–far threshold $d_\theta = 5.0$, and applies targeted near- and far-field interventions. E4 uses the aligned action $a_-(s)$ as directionally useful local negative feedback and the remaining contour actions as additional far-field pressure. The experiment-specific coefficients, training horizons, seeds, taper calibrations, and terminal criteria are reported separately with their corresponding protocols.

C.2 Controlled Categorical Environment D-U1

D-U1 denotes the controlled categorical environment family used to study persistent negative suppression, probability-support boundaries, and shared-semantic generalization. Its E6 instantiation uses a fixed catalogue of 64 unordered actions, each associated with a four-dimensional unit semantic vector. Action identifiers are randomly permuted for every seed, so numerical action indices carry no geometric ordering.

For each seed, we independently sample 2048 training contexts and 2048 test contexts from $s \sim \mathcal{N}(0, I_6)$. The two splits follow the same context distribution, and test contexts never participate in optimization. Results on this split are therefore reported as held-out-context or unseen-state generalization rather than OOD generalization.

Semantic action catalogue. Let $\{e_i\}_{i=1}^{K} \subset \mathbb{R}^4$, with $K = 64$, denote the unit reward embeddings of the categorical actions. They are sampled once per seed and randomly assigned to action identifiers. For each context, two fixed linear maps generate a demonstrated semantic target and a candidate improvement direction:

$$\begin{aligned} t_+(s) &= \mathrm{unit}\big(W_+^\top s\big)\,, \\ \widetilde{u}(s) &= W_u^\top s - \big\langle W_u^\top s, t_+(s)\big\rangle\, t_+(s), \\ u(s) &= \mathrm{unit}\big(\widetilde{u}(s)\big)\,. \end{aligned} \tag{160}$$

The hidden optimal target and the local negative target are

$$\begin{aligned} t_\star(s) &= \mathrm{unit}(t_+(s) + \delta u(s))\,, \\ t_-(s) &= \mathrm{unit}(t_+(s) - \delta u(s))\,, \end{aligned} \qquad \delta = 0.45. \tag{161}$$

Reward and action roles. The ground-truth reward of action i is determined by its semantic alignment with the hidden target:

$$R(s, i) = \frac{1}{2}\left(1 + t_\star(s)^\top e_i\right). \tag{162}$$

The hidden optimal action is

$$i_\star(s) = \arg\max_i t_\star(s)^\top e_i. \tag{163}$$

It is excluded from all positive demonstrations.

For each context, the four positive actions are the available actions with the largest similarity to $t_+(s)$. After excluding the hidden optimum and positive actions, the local negative action is the action most aligned with $t_-(s)$. The four far-negative actions are selected by their alignment with $-t_+(s)$. Thus,

$$\begin{aligned} \mathcal{P}(s) &= \mathrm{Top4}_{i \neq i_\star(s)}\ t_+(s)^\top e_i, \\ i_{\mathrm{local}}(s) &= \arg\max_{i \notin \{i_\star(s)\} \cup \mathcal{P}(s)} t_-(s)^\top e_i, \\ \mathcal{F}(s) &= \mathrm{Top4}_{i \notin \{i_\star(s), i_{\mathrm{local}}(s)\} \cup \mathcal{P}(s)} \left[-t_+(s)^\top e_i\right]. \end{aligned} \tag{164}$$

The positive and negative advantages are fixed throughout training:

$$\widehat{A}(s, i) = \begin{cases} +1, & i \in \mathcal{P}(s), \\ -1, & i = i_{\mathrm{local}}(s) \text{ or } i \in \mathcal{F}(s). \end{cases} \tag{165}$$

Consequently, differences between local and far-negative updates cannot be attributed to different advantage magnitudes.

Shared-semantic categorical policy. The policy uses a two-layer tanh MLP with 64 hidden units per layer to map the context to a unit semantic direction $q_\theta(s) \in \mathbb{R}^4$. Let $\widetilde{e}_i$ denote the action embedding used by the policy. The categorical logits and probabilities are

$$\begin{aligned} \ell_{\theta,i}(s) &= \kappa_\theta(s) q_\theta(s)^\top \widetilde{e}_i, \\ \pi_\theta(i \mid s) &= \frac{\exp\big(\ell_{\theta,i}(s)\big)}{\sum_{j=1}^{K} \exp\big(\ell_{\theta,j}(s)\big)}. \end{aligned} \tag{166}$$

The concentration is either fixed at $\kappa_\theta(s) = 8$ or learned as

$$\kappa_\theta(s) = \mathrm{softplus}(c_\theta(s)) + 0.05, \tag{167}$$

with initial value 8 and no upper clamp.

In the semantically aligned condition, $\widetilde{e}_i = e_i$. In the shuffled control, the same policy embeddings are randomly permuted across action identifiers while the reward embeddings and training samples remain unchanged. This intervention preserves the catalogue size and categorical parameterization but breaks the correspondence between policy-side similarity and ground-truth semantic utility.

Evaluation and failure events. We evaluate expected semantic reward, probability assigned to the hidden optimal action, entropy, effective support, and normalized semantic extrapolation on both training and held-out contexts. Task-performance collapse, effective-support or concentration-boundary events, and NaN/Inf numerical failure are reported separately. A policy can therefore retain high reward while still approaching a support boundary; the two outcomes are not treated as equivalent.

Shared use across categorical protocols. E6-A fixes the concentration and scans the strength of the local negative update to identify the transition from the positive-only ceiling to useful semantic extrapolation and then to performance reversal. E6-B learns the concentration and applies near/far causal interventions and matched controls. E6-C compares aligned and shuffled policy embeddings to determine whether improvements on the hidden action require the specified semantic correspondence.

E5 serves a distinct but complementary role. Its direct-softmax diagnostic and categorical causal reconstruction isolate persistent surprisal growth, logit-gap growth, and simplex-boundary suppression under repeated negative updates. It shares the fixed-advantage and near/far intervention logic of D-U1, but it does not use the 64-action shared-semantic catalogue and does not support the E6 semantic generalization claim.

C.3 D4RL Hopper Datasets

Hopper is a continuous-control task with an 11-dimensional observation space and a three-dimensional bounded action space. We use the D4RL `hopper-medium`, `hopper-medium-replay`, and `hopper-medium-expert` datasets. The medium dataset contains trajectories collected from a partially trained policy, medium-replay contains the corresponding replay buffer collected throughout training, and medium-expert combines medium and expert behavior. Hopper-medium-replay is used for the primary mechanism diagnostic because its broader behavioral coverage provides a wider range of learner-relative distances, while all three Hopper datasets are retained for the task-level performance evaluation.

C.4 Countdown Task and Puzzle Dataset

Countdown is an arithmetic generation task in which the policy receives a multiset of integers and a target value, and must generate an arithmetic expression that evaluates exactly to the target. A valid output must use every supplied number exactly once and may contain only addition, subtraction, multiplication, division, and parentheses.

Let $x = (\mathcal{N}, T)$ denote a puzzle, where $\mathcal{N}$ is the supplied multiset of numbers and T is the target. The policy action is a complete token sequence y representing the generated expression. Task success is determined by a deterministic verifier:

$$R(x, y) = \mathbf{1}\begin{bmatrix} y \text{ is syntactically valid, uses } \mathcal{N} \text{ exactly once,} \\ \text{and evaluates to } T \end{bmatrix}. \tag{168}$$

Because multiple expressions can solve the same puzzle, evaluation is based on verifier success rather than exact string matching.

The puzzle corpus is procedurally generated from valid arithmetic expressions and divided into disjoint training, validation, and test sets. Duplicate puzzles with the same number multiset and target are removed across the corpus. Validation puzzles are used for checkpoint selection, while the test split is reserved for final evaluation.

Countdown provides an external discrete-action setting with autoregressive sequence generation and shared Transformer parameters. For the diagnostic in Figure 1, unsuccessful responses are verifier-matched and grouped by current mean-token surprisal.

C.5 Training and Evaluation Protocols

D Controlled Experimental Protocols

This section specifies the interventions used in Sections 6.3 and 6.4. The definitions of C-U1 and D-U1 are given in Appendix C. Network architectures, optimizer settings, training budgets, seeds, and evaluation frequencies are reported in the protocol-specific subsections below.

D.1 Shared Notation and Experimental Invariants

For a negative sample $z = (s, a)$, define its raw repulsive update as

$$g_\theta^-(z) = -\widehat{A}^-(z)\nabla_\theta \log \pi_\theta(a \mid s), \qquad \widehat{A}^-(z) = \max\{-\widehat{A}(z), 0\}. \tag{169}$$

Learner-relative remoteness is measured by

$$D_\theta(z) = -\log \pi_\theta(a \mid s), \tag{170}$$

or by its equivalent standardized Gaussian component when additive density constants are irrelevant.

Unless explicitly stated otherwise, all branches of one controlled comparison share:

(1) the same training and evaluation contexts;
(2) the same positive and negative samples;
(3) the same fixed rewards or advantage labels;
(4) the same initial policy parameters;
(5) the same minibatch order, optimizer, and training horizon;
(6) the same evaluation checkpoints and terminal classification rules.

Only the variable named by the intervention is changed. Actions generated under the same context are aggregated at the context level and are not treated as independent contexts for statistical inference.

D.2 Source-Isolation Protocol

Purpose. The source-isolation experiment asks whether policy-relative remoteness changes negative-update magnitude when sample quality is held fixed. It identifies the source of a per-sample difference, not its causal effect on training.

Continuous product-manifold construction. For each C-U1 context, we select multiple negative actions from a common reward contour:

$$R(s, a_i) = R(s, a_j), \qquad \widehat{A}(s, a_i) = \widehat{A}(s, a_j) < 0, \tag{171}$$

while ensuring that

$$D_\theta(s, a_i) \neq D_\theta(s, a_j). \tag{172}$$

The reward coordinate and the policy-distance coordinate are therefore independently controlled. This exact counterfactual comparison is the reason for using a synthetic product-manifold construction: an external offline dataset generally cannot provide two actions with precisely matched task quality and independently prescribed distance from the same current policy.

The comparison is performed both at policy initialization and after positive-only pretraining. No negative sample used as a probe is applied as an optimizer update during this protocol. Let $u_\theta(s)$ denote the policy-output coordinate used by the controlled probe, e.g., fixed-covariance Gaussian mean coordinates in C-U1 and direct logits in D-U1. For every probe, we record

$$\begin{aligned} I_{\text{score}}(z) &= \|\nabla_u \log \pi_u(a \mid s)\|_2, \\ I_{\text{sample}}(z) &= |\widehat{A}(z)|\, I_{\text{score}}(z). \end{aligned} \tag{173}$$

Because $|\widehat{A}|$ is identical across the matched actions, differences in I_{sample} arise from policy-score geometry rather than advantage magnitude.

For neural implementations, we additionally record the full parameter-gradient norm and the aggregate gradient obtained from samples in the same remoteness range. These implementation-level actor-gradient diagnostics are used as transfer checks, not as theorem-level score measures.

We report near-to-far ratios of remoteness, score norm, single-sample gradient norm, aggregate gradient norm, and gradient-direction coherence. Ratios are first computed within each context and seed and are then aggregated across the prespecified seeds.

Categorical counterpart. D-U1 uses unordered categorical actions with identical fixed negative labels. The comparison varies the current probability assigned to the negative action while preserving its reward role and advantage:

$$\widehat{A}(s, i) = \widehat{A}(s, j) < 0, \qquad \pi_\theta(i \mid s) \neq \pi_\theta(j \mid s). \tag{174}$$

For a direct categorical logit parameterization, the score norm is bounded and need not grow without limit as probability decreases. The relevant far-field effect is instead persistence: the negative update continues to increase the selected action's logit gap even after its probability has become small. We therefore report probability, surprisal, logit gap, score norm, and cumulative suppression over repeated updates. This distinguishes categorical support suppression from the unbounded Gaussian-distance amplification measured in C-U1.

D.3 Causal-Transmission Protocol

Purpose. The source-isolation protocol identifies the source of abnormal update magnitude but not its effect on task behavior. Causal transmission is tested in the nonlinear Gaussian protocol, where the policy mean and variance evolve jointly, and in the categorical support reconstruction of D-U1.

Dynamic near–far partition. At each intervention step, negative samples are partitioned using their current-policy remoteness. Let $\mathcal{N}_\theta$ and $\mathcal{F}_\theta$ denote the prespecified near- and far-field sets. Their membership is recomputed as the policy evolves rather than frozen at initialization. The thresholds or quantiles are fixed before training and specified with the corresponding environment.

Interventions. For a negative sample z, the following multipliers are applied to $g_\theta^-(z)$:

$$\begin{aligned} w_{\text{uncontrolled}}(z) &= 1, \\ w_{\text{positive}}(z) &= 0, \\ w_{\text{near-zero}}(z) &= \mathbf{1}[z \notin \mathcal{N}_\theta], \\ w_{\text{far-zero}}(z) &= \mathbf{1}[z \notin \mathcal{F}_\theta]. \end{aligned} \tag{175}$$

The far-cap intervention preserves the direction of a far-field update but limits its detached magnitude to a reference level estimated from near-field negative samples:

$$w_{\text{far-cap}}(z) = \begin{cases} \min\left\{1, \dfrac{C_{\text{near}}}{I_{\text{sample}}(z) + \epsilon}\right\}, & z \in \mathcal{F}_\theta, \\ 1, & \text{otherwise,} \end{cases} \tag{176}$$

where C_{near} is the prespecified near-field update-magnitude reference.

To distinguish selective intervention from a generic decrease in negative optimization, the global-matched control applies one scalar to every negative sample:

$$w_{\text{global}}(z) = \alpha_{\text{match}}. \tag{177}$$

The scalar is chosen using the training batch so that its detached aggregate negative-update magnitude matches the corresponding selective intervention:

$$\alpha_{\text{match}} \sum_{z \in \mathcal{B}^-} I_{\text{sample}}(z) = \sum_{z \in \mathcal{B}^-} w_{\text{selective}}(z) I_{\text{sample}}(z). \tag{178}$$

The matching coefficient is not differentiated through.

Causal contrasts. The primary causal contrast is between removing near-field and far-field negative updates. If Near-zero leaves the dynamics unchanged while Far-zero or Far-cap alters them, the effect cannot be attributed to negative feedback in general. The global-matched branch quantifies how much of the change follows from reducing total negative-update magnitude and how much requires selecting samples by remoteness.

Temporal and terminal measurements. We record policy displacement, reward, negative-gradient magnitude, mean dynamics and, for learnable-variance runs, scale dynamics, entropy or effective support, and the first time at which each prespecified event occurs. The analysis checks whether far-field update magnitude rises before policy drift and whether drift precedes task-performance loss.

Terminal outcomes are classified separately as:

(1) **task-performance collapse**: a sustained loss of the prespecified task metric while optimization remains finite;
(2) **support or variance-boundary event**: categorical effective support or Gaussian variance crosses its prespecified boundary without requiring NaN or Inf;
(3) **numerical collapse**: a parameter, loss, gradient, or model output becomes NaN or Inf.

A boundary event is not automatically counted as task-performance collapse, and finite execution is not automatically counted as convergence.

D.4 Controlled Negative-Strength Sweep

The controlled negative-strength sweep varies only the effective negative-repulsion strength q/p, where p denotes the aggregate positive-attraction budget and q denotes the aggregate negative-repulsion budget. All sweep arms share the same data, fixed labels, initialization, optimizer, training horizon, and event thresholds. The positive-only baseline corresponds to $q/p = 0$.

Held-out-context reward evaluates unseen-state generalization within the C-U1 controlled protocol. Policy shift is the normalized displacement of the learned policy from the positive-only target and visualizes how far negative feedback pushes the policy beyond the imitation solution. Task-performance collapse, support or variance-boundary events, and NaN/Inf numerical failures are reported as separate event types.

D.5 Controlled Taper Comparison

Controlled utility construction. In C-U1, the hidden optimum $a_\star(s)$ lies beyond the center of the positive demonstrations $a_+(s)$. A local negative anchor is placed on the opposite side of $a_+(s)$, so repulsion from this anchor points toward $a_\star(s)$. Additional negative actions are placed on matched reward contours at larger policy-relative distances. Increasing negative strength therefore produces three observable regimes: positive-only under-extrapolation, useful local extrapolation, and far-field over-extrapolation or instability.

D-U1 implements the same logic with unordered semantic actions. The hidden optimal target lies beyond the demonstrated positive direction, while the local negative target lies in the opposite semantic direction. Repelling the local negative action can therefore improve the shared representation toward the hidden optimal action. Additional remote negative actions create support pressure without changing their fixed negative label. Evaluation on independently sampled contexts is reported as held-out-context generalization.

Let

$$x(D) = \left[\frac{D - \tau}{c}\right]_+, \qquad r_x(D) = \sqrt{x(D)}.$$

The Linear and Quadratic names refer to polynomial order in the radialized coordinate $r_\theta = \sqrt{x_\theta}$, rather than in the squared-remoteness coordinate x_θ.

Weighting rules. For negative-sample remoteness D, the compared rules are

$$\begin{aligned} \omega_{\text{positive}}(D) &= 0, \\ \omega_{\text{uncontrolled}}(D) &= 1, \\ \omega_{\text{global}}(D) &= \alpha, \\ \omega_{\text{hard}}(D) &= \mathbf{1}[D \le \tau], \\ \omega_{\text{Rec-L}}(D) &= \frac{1}{1+\lambda r_x(D)}, \\ \omega_{\text{Rec-Q}}(D) &= \frac{1}{1+\lambda r_x(D)^2}, \\ \omega_{\text{exp}}(D) &= \exp\{-\lambda r_x(D)^2\}. \end{aligned} \tag{179}$$

All weighting values are computed from detached remoteness statistics; the policy cannot reduce its loss by differentiating through the taper.

Near-field-retention matching. Let $\mathcal{N}$ denote the prespecified useful near-field set and define its retained first-order update magnitude as

$$\rho_{\text{near}}(\omega) = \frac{\sum_{z\in\mathcal{N}} \omega(D_\theta(z)) I_{\text{sample}}(z)}{\sum_{z\in\mathcal{N}} I_{\text{sample}}(z)}. \tag{180}$$

Taper parameters are calibrated on training contexts so that the compared selective rules retain the same target near-field fraction, up to the predetermined tolerance. Parameters are then frozen before evaluation. This comparison isolates how the rules treat the far-field tail after preserving comparable local utility.

Aggregate-budget matching. For a weighting function ω, define its detached negative-update budget as

$$B(\omega) = \sum_{z\in\mathcal{B}^-} \omega(D_\theta(z)) I_{\text{sample}}(z). \tag{181}$$

The budget-matched global coefficient is

$$\alpha_{\text{budget}} = \frac{B(\omega)}{B(\omega_{\text{uncontrolled}})}. \tag{182}$$

This control applies the same total first-order negative-update budget without using remoteness. A selective taper can therefore be credited only for improvements that remain after comparison with its budget-matched global baseline.

Measurements. Each method is evaluated over the full prespecified training horizon. We report task performance, held-out-context generalization, distance from the hidden optimum, policy displacement, entropy or variance, effective support, near-field retention, aggregate negative-update budget, and all three terminal-event categories. Method comparisons use the same seeds and paired statistics.

D.6 External Taper Validation

Scientific scope. External taper experiments test whether the controlled weighting principle transfers to complex policies. They do not reproduce the exact counterfactual geometry of C-U1 or D-U1 and therefore do not replace the controlled source and causal experiments.

Hopper. All Hopper taper methods branch from the same actor initialization and use the same offline transitions and advantage labels. For an action with inverse-squash coordinate u, Gaussian mean $\mu_\theta(s)$, and diagonal scale $\sigma_\theta(s)$, the remoteness statistic is

$$D_{\text{Hopper}}(s,a) = \frac{1}{2}\sum_j \left(\frac{u_j - \mu_{\theta,j}(s)}{\sigma_{\theta,j}(s)}\right)^2. \tag{183}$$

The additive Gaussian normalization term is omitted because it does not change the ordering used by the taper. Remoteness is recomputed under the current actor and detached before constructing the negative weight. The mechanism protocol with frozen advantages is kept separate from the standard offline-RL performance protocol; the former studies transfer of the far-field control mechanism, whereas the latter reports final environment return.

Countdown. The Countdown offline dataset is generated once from a frozen SFT policy. For each puzzle, it contains a verified positive response, a near negative response with low mean token surprisal, and a far negative response with high mean token surprisal. Mean token surprisal is

$$D_{\text{Countdown}}(x,y) = -\frac{1}{|y|}\sum_{t=1}^{|y|} \log \pi_\theta\left(y_t \mid x, y_{<t}\right). \tag{184}$$

During method training, the responses remain fixed but their surprisal is recomputed under the current policy. The sequence-level taper weight is computed from the detached current surprisal and multiplies the negative sequence objective.

All Countdown methods start from the same SFT checkpoint and use the same positive, near-negative, and far-negative responses, mixture coefficients, validation set, and maximum training budget. Checkpoint selection uses the registered validation task metric, and final verifier success is measured only on the held-out test puzzles.

Comparisons and reporting. The common external comparisons are Positive-only, Uncontrolled, Global scaling, and exponential DRPO. Domain-specific baselines are reported in the overall-performance section rather than treated as controlled taper variants. Hopper reports normalized return and policy statistics; Countdown reports greedy verifier success, pass@k, valid expression rate, surprisal, and support statistics. Task-performance, boundary, and NaN/Inf outcomes remain separate in both domains.

D.7 Shared Objectives and Negative-Update Controls

This section defines the protocol controls and remoteness-aware weighting rules shared by the D4RL and Countdown experiments. Their task-specific remoteness signals, data construction, and optimization protocols are described separately in Appendices D.8 and D.9.

Shared signed objective. Let z denote an offline transition or response, let $A(z)$ denote its fixed signed learning signal during an actor update, and let $D_\theta(z)$ denote its learner-relative remoteness under the current policy. We write

$$A^+(z) = \max\{A(z), 0\}, \qquad A^-(z) = \max\{-A(z), 0\}.$$

The shared family of actor objectives is

$$\mathcal{J}_\omega(\theta) = \mathbb{E}_{z\sim\mathcal{B}}\left[A^+(z)\log\pi_\theta(z) - \omega(D_\theta(z))\,A^-(z)\log\pi_\theta(z)\right],$$

where positive updates retain unit weight and $\omega(D)$ controls only the negative branch. During each actor update, the advantage and remoteness-derived weight are treated as fixed coefficients; gradients are not propagated through the weighting rule.

For the remoteness-aware methods, define the normalized excess remoteness

$$x_\theta(z) = \left[\frac{D_\theta(z)-\tau}{c}\right]_+,$$

where τ is the near-field threshold, $c > 0$ is the task-specific scale, and $[u]_+ = \max\{u, 0\}$. Consequently, samples inside the near-field region $D_\theta(z) \le \tau$ retain their full negative weight.

Positive-only. Positive-only removes all negative updates:

$$\omega_{\text{pos}}(D) = 0.$$

It therefore provides an extreme no-repulsion reference. In D4RL it retains only transitions with positive actor advantages; in Countdown it retains only verified successful responses under the corresponding binary learning signal.

Global-α. Global-α applies the same attenuation to every negative sample:

$$\omega_{\text{global}}(D) = \alpha, \qquad 0 \le \alpha \le 1.$$

This control tests whether performance improvements can be explained by a global reduction in negative-update magnitude, without using learner-relative remoteness.

Reciprocal-Linear and Reciprocal-Quadratic. For the remoteness-aware methods, define the normalized excess remoteness

$$x_\theta(z) = \left[\frac{D_\theta(z)-\tau}{c}\right]_+,$$

and its radialized coordinate

$$r_\theta(z) = \sqrt{x_\theta(z)}.$$

For Gaussian policies, r_θ is asymptotically proportional to standardized distance in the far field; for sequence policies, it is a surprisal-derived radial coordinate rather than a geometric distance in the discrete action space. The two reciprocal controls use the same remoteness signal and near-field threshold as DRPO, but differ in their far-field decay:

$$\omega_{\text{Rec-L}}(D) = \frac{1}{1+\lambda r_\theta(z)},$$

and

$$\omega_{\text{Rec-Q}}(D) = \frac{1}{1+\lambda r_\theta(z)^2} = \frac{1}{1+\lambda x_\theta(z)}.$$

Reciprocal-Linear has the slowest far-field attenuation. The quadratic variant decays more rapidly but retains a polynomial tail. These controls isolate whether the benefit comes merely from using a remoteness signal or specifically from the decay profile selected by DRPO.

DRPO.. DRPO uses an exponential remoteness taper:

$$\omega_{\text{DRPO}}(D) = \exp\{-\lambda x_\theta(z)\}.$$

The weight equals one throughout the near-field region and then decays exponentially once remoteness exceeds τ. Thus, DRPO preserves local negative feedback while suppressing the remote negative tail more rapidly than the reciprocal alternatives.

Hyperparameter selection. Within each task, τ, c, λ, and the Global-α coefficient are selected using the registered validation protocol only. Test-task performance is not used for hyperparameter selection. All protocol-matched methods share the same candidate budgets and checkpoint-selection rule. The task-specific search spaces and selected values are reported in Appendices D.8 and D.9.

D.8 D4RL Baselines and Experimental Protocol

The definitions of Positive-only, Global-α, Reciprocal-Linear, Reciprocal-Quadratic, and DRPO follow Appendix D.7. This section specifies only the D4RL-specific baselines, remoteness construction, training protocol, and evaluation procedure.

Tasks and datasets. We evaluate HalfCheetah, Hopper, and Walker2d on the medium, medium-replay, and medium-expert datasets, producing the nine tasks reported in Table 1. The same fixed offline dataset is used by every protocol-matched method within a task.

Published offline-RL baselines. CQL, TD3+BC, and IQL are included as published reference results. These values are not protocol-matched reruns and are therefore marked by † in Table 1. The main table reports their published task-level means. Their original uncertainty statistics and exact score provenance are recorded separately from the protocol-matched DRPO experiments.

Gaussian-policy remoteness. For an action $a \in \mathbb{R}^{d_a}$ and current Gaussian policy with mean $\mu_\theta(s)$ and standard deviation $\sigma_\theta(s)$, we use the following implementation-level standardized remoteness statistic

$$D_{\text{D4RL}}(s,a) = \frac{1}{d_a}\sum_{j=1}^{d_a}\left(\frac{a_j-\mu_{\theta,j}(s)}{\sigma_{\theta,j}(s)}\right)^2.$$

This dimension-normalized convention differs from Equation (183) only by a fixed task-level scale. The associated threshold and scale parameters absorb that constant, and each protocol keeps its convention fixed across all compared methods.

When the actor uses a transformed action distribution, this quantity is computed in the policy's native Gaussian coordinates. The resulting remoteness is used only to determine the negative-update coefficient; the distance term is detached during the actor update.

Protocol-matched actor comparison. Positive-only, Global-α, Reciprocal-Linear, Reciprocal-Quadratic, and DRPO share the same offline dataset, critic checkpoint or critic-training protocol, actor initialization, network architecture, optimizer, minibatch schedule, actor-update budget, evaluation schedule, and random seeds. They differ only in the negative-update weighting rule defined in Appendix D.7.

Table 3: Registered configuration for the D4RL performance experiments.

Configuration	Registered value
Actor-update budget	**[registered value]**
Evaluation interval	**[registered value]**
Evaluation episodes	**[registered value]**
Random seeds	**[registered seed set]**
Near-field threshold candidates, τ	**[registered grid]**
Remoteness-scale candidates, c	**[registered grid]**
Taper-strength candidates, λ	**[registered grid]**
Global-scaling candidates, α	**[registered grid]**

The common actor advantage construction is held fixed across these methods. No method is allowed to change the critic, relabel samples, or alter the data distribution in order to improve its actor result.

Hyperparameters. The registered training and selection configuration is summarized in Table 3. All entries must be populated directly from the frozen experiment configuration rather than reconstructed after observing the test results.

Evaluation and uncertainty. Task performance is reported using the standard D4RL normalized return. For every protocol-matched method, the main table reports the mean over the registered seeds, while seed-level values and uncertainty statistics are reported in the appendix. The D4RL-9 total is the sum of the nine task-level means.

Task-performance collapse is assessed from normalized return and training trajectories. Variance or support-boundary events are reported separately from task performance. NaN/Inf numerical failures are reported as a third outcome and are not merged with either low return or policy-boundary events. Any claim concerning convergence, collapse, or method ranking is based on the registered terminal audit rather than an intermediate checkpoint.

D.9 Countdown Baselines and Experimental Protocol

Positive-only, Global-α, Reciprocal-Linear, Reciprocal-Quadratic, and DRPO follow the shared definitions in Appendix D.7. This section defines the two published off-policy baselines and the Countdown-specific response-bank, remoteness, training, and evaluation protocols.

AsymRE.. Asymmetric REINFORCE defines the response-level advantage using a tunable reward baseline V:

$$A_{\mathrm{AsymRE}}(x, y) = r(x, y) - V,$$

and optimizes

$$\mathcal{J}_{\mathrm{AsymRE}}(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{B}}\left[\left(r(x, y) - V\right)\log \pi_\theta(y \mid x)\right].$$

For a binary verifier reward, lowering V emphasizes successful responses, whereas increasing V strengthens suppression of failed responses. The baseline is selected using the validation protocol and is held fixed within each run.

Global-α and AsymRE remain separate comparisons because they operate on different underlying learning signals. Global-α scales the negative part of the common precomputed signed advantage, whereas AsymRE constructs its signed advantage directly from the raw verifier reward and the published baseline parameterization. Under binary rewards and simplified normalization they are closely related, but the formal comparison preserves their native objectives and tuning rules.

TOPR.. Let $\mu(y \mid x)$ denote the frozen response-generating policy and

$$\rho_\theta(x, y) = \frac{\pi_\theta(y \mid x)}{\mu(y \mid x)}$$

the sequence-level importance ratio. Let $R(x, y)$ denote the signed reward used by TOPR and define

$$\mathcal{T}^+ = \{(x, y) : R(x, y) \geq 0\}, \qquad \mathcal{T}^- = \{(x, y) : R(x, y) < 0\}.$$

We implement canonical TOPR, whose expected update is

$$\begin{aligned} g_{\mathrm{TOPR}}(\theta) = \mathbb{E}_{(x,y)\sim\mu}\Big[&\mathbf{1}\{(x, y) \in \mathcal{T}^+\} R(x, y) \\ &\nabla_\theta \log \pi_\theta(y \mid x)\Big] \\ + \mathbb{E}_{(x,y)\sim\mu}\Big[&\mathbf{1}\{(x, y) \in \mathcal{T}^-\} \min\{\rho_\theta(x, y), 1\} R(x, y) \\ &\nabla_\theta \log \pi_\theta(y \mid x)\Big]. \end{aligned}$$

Thus, positive responses receive the SFT-style unit coefficient, while negative responses are attenuated using a truncated importance ratio. The behavior-policy log probability required by the ratio is stored when the frozen response bank is constructed.

Countdown remoteness. For a prompt x and completion $y = (y_1, \ldots, y_T)$, learner-relative remoteness is the current mean completion-token surprisal

$$D_{\mathrm{CD}}(x, y) = -\frac{1}{T}\sum_{t=1}^{T} \log \pi_\theta\left(y_t \mid x, y_{<t}\right).$$

Only completion tokens are included; prompt tokens and padding positions are excluded. Mean rather than summed surprisal is used so that response length does not mechanically determine remoteness. The remoteness score is detached before applying the shared taper functions.

Frozen response-bank construction. The response bank is generated before policy optimization and then held fixed. Each record contains the puzzle, generated response, verifier outcome, response length, and the behavior-policy log probability required by methods that use importance ratios. The same bank and training split are supplied to every protocol-matched method.

A response is marked successful only when it forms a valid arithmetic expression, uses the supplied numbers according to the task rules, and evaluates to the target. Invalid syntax, illegal number use, and incorrect arithmetic results remain distinct verifier outcomes

Table 4: Registered configuration for the Countdown performance experiments.

Configuration	Registered value
Optimizer-step budget	**[registered value]**
Batch size	**[registered value]**
Learning-rate schedule	**[registered value]**
Random seeds	**[registered seed set]**
AsymRE baseline candidates, V	**[registered grid]**
Near-field threshold candidates, τ	**[registered grid]**
Remoteness-scale candidates, c	**[registered grid]**
Taper-strength candidates, λ	**[registered grid]**
Global-scaling candidates, α	**[registered grid]**
Number of samples for pass@k	**[registered value]**

in the stored diagnostics, even when they share the same binary training reward.

Training and checkpoint selection. All methods start from the same SFT checkpoint and share the optimizer, learning-rate schedule, minibatch construction, number of optimizer steps, validation split, test puzzles, and random seeds. Hyperparameters are selected using the registered validation rule. The held-out test set is evaluated only after checkpoint selection.

The registered training and selection configuration is summarized in Table 4. All entries must be populated directly from the frozen experiment configuration rather than reconstructed after observing the test results.

Verifier-based evaluation. Greedy verifier success is the fraction of held-out puzzles solved by greedy decoding. Pass@k is the fraction solved by at least one of the registered k sampled responses. Valid-expression rate is the fraction of generated responses that satisfy the parser and number-use constraints, irrespective of whether they reach the target.

These three metrics are reported separately. In particular, a higher verifier success rate is not attributed to improved reasoning when it is accompanied by a deterioration in expression validity.

Task-performance degradation, token- or sequence-support boundary events, and NaN/Inf numerical failures are audited independently. Claims about stability or final method ranking use the registered terminal checkpoint audit rather than finite-step pilot behavior.

D.10 Countdown Taper-Coefficient Sensitivity

Figure 5 reports the complete stored terminal Pass@8 coefficient response for the Countdown-0.5B taper study under the fixed 1,200-step budget. Each point is the mean of the two paired development seeds, and each error bar spans the corresponding pair of seed values. The horizontal reference is the Positive-only terminal mean of 14.3%.

The exponential response rises above Positive-only over an intermediate coefficient region, reaches a maximum two-seed terminal mean of 16.6%, and returns toward the Positive-only level at the largest evaluated coefficients. Reciprocal-Linear increases toward a broad plateau but remains below Positive-only, with a maximum terminal mean of 13.9%. Reciprocal-Quadratic reaches 15.0% at $\lambda = 24$ and then fluctuates around a broad reference-level plateau through $\lambda = 1279$. These complete curves show that the observed difference is a response-shape effect rather than a comparison of isolated best points. Because this study uses two paired development seeds and a fixed horizon, it supports a bounded tail-shape sensitivity claim, not convergence, statistical significance, or a universal method ranking.

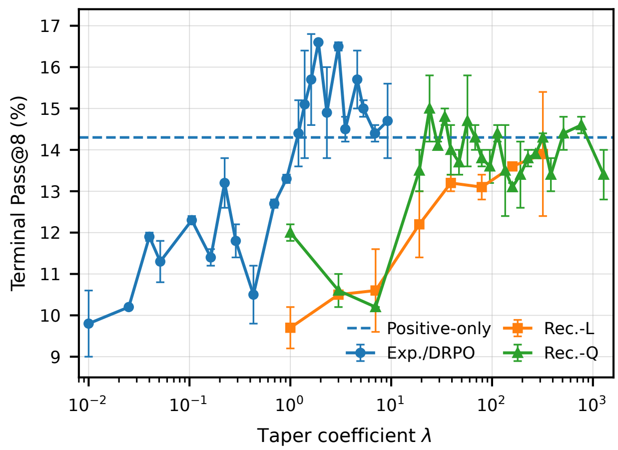

Figure 5: Countdown-0.5B taper-coefficient response. Terminal Pass@8 is shown against the taper coefficient λ under the fixed 1,200-step budget. Curves report two-seed means; error bars span the paired seed values, and the dashed horizontal line denotes Positive-only. Exp./DRPO is shown over $\lambda \in [0.01, 9.2103]$, Reciprocal-Linear through 319, and Reciprocal-Quadratic through 1279.